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D4.2 – Decision Support Platform – report and demonstration v1

WP4: Platform for Decision Making Processes for a Disaster Resilience Society

30 October 2024 (M25)



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Acronyms

Acronym	Description
C2IMPRESS	Co-Creative Improved Understanding and Awareness of Multi-Hazard Risks for Disaster Resilient Society
GA	Grant Agreement
DSS	Decision Support System
MCA	Multi-criteria Analysis
DMOP	Disaster Monitoring & Observation Portal
DPSIR	Driver, Pressure, State, Impact, Response
PESTLE	Political, Economic, Social, Technological, Legal, and Environmental
CUI	Common User Interface
API	Application Programming Interface
VKG	Virtual Knowledge Graph
EWS	Early Warning System
SMCDA	Spatial Multi Criteria Decision Analysis
HBM	Human Behavior Modeling
ABM	Agent-Based Modeling
PPCP	Public-Private-Civic Partnership
SoS4MHRIN	System-of-Systems for Multi-hazard Risk Intelligent Network

1 Introduction

The C2IMPRESS decision support platform aims to deliver a robust, flexible, and interactive suite of microservices² to provide access to simulation tools, Early Warning Systems (EWS), methods/guidelines, decision analysis, and support tools to optimise the operational framework of dynamic risk, resilience, and climate change adaptation. The platform, compared to any existing systems or approaches, will support the end-users through: (i) improved accuracy, consistency and interoperability of key input data through a knowledge graph, (ii) improved understanding of the long-term climatic projections and related hazards, scenarios, and associated uncertainties (temporal and spatial scale), (iii) precise understanding of a particular hazard and its compound and/ or cascading effects, (iv) accurate extent and degree of ecosystem vulnerability and exposure due to multi-hazards, (v) climate change (CC) scenario development and prioritisation using decision support system, (vi) CC impact assessment and management at the local-regional and multilevel scale, according to the requirements of policy and regulatory frameworks, and to support to revise existing risk assessment frameworks, (vii) specific guidelines for multi-level governance structure formulation for climate change adaptation (CCA) policy implementation, evaluation or reassessment and (viii) a citizen engagement framework for to ensure engagement and participation of the end users.

The platform integrates the microservices that have been developed and are being developed in different work packages under the scope of the C2IMPRESS project. Each of the tools has its own functionality and specific target end-user. Most of the tools of the C2IMPRESS project are based on the existing tools and knowledge prior to the start of the project (TRL3-4) and targeting to achieve TRL5-7, depending on the scope and opportunities. This platform is modular and works as a one-stop mall for the tools and methods used in this project.

The D4.2 is an initial version of the Platform description and we structured the sections are as follows:

Chapter 2 explain the technical development of the platform. It contains how are the tools integrated.

Chapter 3 provides a brief summary of each tool, their end user, the decision support that each tool will provide and the procedure of integration and interconnection the tools.

Chapter 4 explains the features and functionalities of the two decision support tools: (a) Multi-criteria decision Analysis, b) DPSIR and PESTLE analysis.

The next steps are stated in Chapter 5.

² Miroservice indicates C2IMPRESS decision support tools, forecasting and EWS, map server, visualisation tools, story map, frameworks



2 Development of the C2IMPRESS Multi-Hazard Decision Support Platform

2.1 Overview

The C2IMPRESS Multi-hazard Decision Support platform, in Task 4.7, is a critical element of the C2IMPRESS project, developed to assist stakeholders in making well-informed decisions regarding disaster risk management, water resource management, environmental risks, and climate change impacts. The DSS functions as an integrated platform that connects a variety of microservices through a centralized user interface, allowing users to retrieve data, assess risks, get early warning, and perform scenario-based analyses.

The system's architecture is structured around a common user interface (CUI) that interacts with an API gateway serving as the backend infrastructure. The API gateway is responsible for managing various services, including authentication, data handling, webhook services, and third-party API integration, enabling efficient communication across the diverse microservices of the C2IMPRESS ecosystem. These microservices, spanning knowledge graphs, early warning systems, and specialized environmental models, work in concert to deliver valuable insights and analytics.

The DSS also incorporates various embedded tools and widgets to enhance data visualization and user interaction. These features, such as iframes, widgets, and embedded links, ensure users have streamlined access to internal and external data resources. By connecting to a centralized database management system, the DSS facilitates efficient data storage and retrieval, supporting databases like PostgreSQL and MongoDB, ensuring interoperability and scalability.

2.2 Development of Common User Interface

The Common User Interface (CUI) is a pivotal component of the Decision Support System (DSS), designed to provide users with a cohesive and streamlined experience for accessing multi-hazard risk management tools.



Figure 2.1: DSS Landing Page



The interface offers an intuitive and responsive platform where users can seamlessly engage with various services, such as hazard forecasting, policy making, social media analytics, Hazard Monitoring etc.

The CUI is engineered to integrate external datasets and microservices through embedded widgets, iframes, and API-driven content, offering a flexible and modular approach to data interaction. Users benefit from a visually clear, structured interface that facilitates seamless data exchange across the platform's various components, enhancing their decision-making process.

Upon successful authentication (Figure 2.1), users are guided to a comprehensive dashboard that is the central hub for accessing decision support tools. These tools are categorized logically, allowing for easy navigation between services such as Early Warning Systems, Simulation Models, and Risk Management Frameworks. The dashboard design prioritizes user efficiency by providing real-time access to critical datasets, forecasting models, and analysis tools—all within a unified platform.

Developed using **Vue.js**, a progressive JavaScript framework known for its flexibility and lightweight nature, the CUI harnesses Vue's core features like component-based architecture and reactive data binding. These fundamentals ensure a dynamic and efficient user experience, making developing and maintaining interactive web applications easy. This approach allows for scalability, ensuring the interface remains responsive and adaptable to future service extensions and updates.

The DSS platform employs a cookie consent mechanism (Figure 2.2) to ensure user privacy and compliance with data protection regulations. Upon initial access to the site, visitors are presented with a clear and concise notice informing them about the use of cookies and their purpose. Visitors have the option to provide explicit consent for the use of cookies or to decline them. By obtaining informed consent, we adhere to data protection best practices and safeguard user privacy.

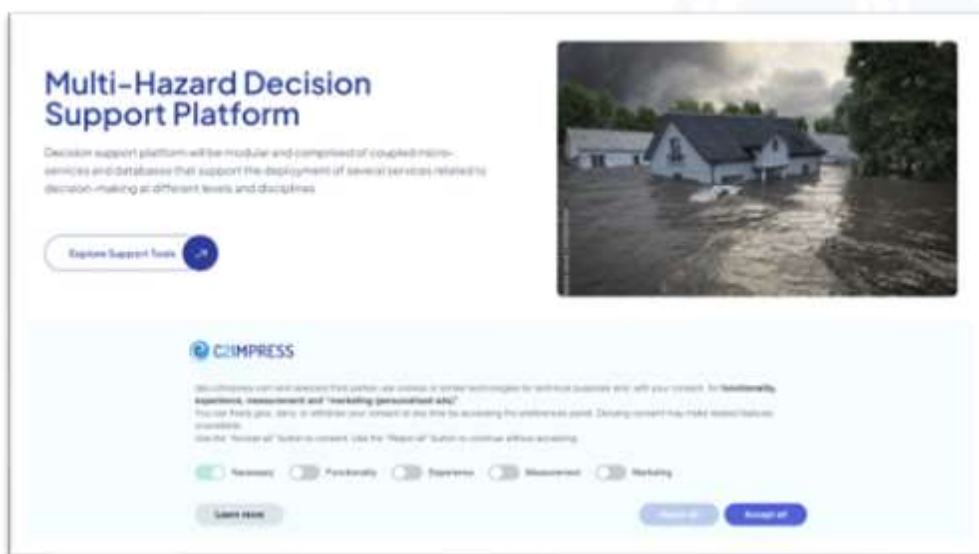


Figure 2.2: DSS Page Cookie Consent Manager

2.3 Microservices

Microservices – also known as microservice architecture – is an architectural style that structures an application as a collection of services that are

- Independently deployable
- Loosely coupled

Services are typically organized around different capabilities. Each service can be owned or managed by an individual team.

Key objectives of microservices are –

- Independence: Microservices can be developed, deployed, and maintained separately.
- Scalability: Each service can be scaled independently to meet demand.
- Decentralization: Each microservices can use different technologies or databases if needed.
- Resilience: If one microservice fails, it doesn't necessarily bring down the entire application.

The main objective and key objectives of this task are to bring all the services under the common user interface (CUI).

2.4 Integration and interconnection of Microservices in DSS Common User Interface (CUI)

The best possible way to connect different microservices to one place is by using an API gateway. It will be a middle point between the CUI and the various microservices. This API gateway can have some internal services to provide a controlled interface between external clients and a microservice-based architecture, improving security, performance and management. For example:

- **Auth service:** This can handle user authentication centrally, allowing users to interact with different microservices in a controlled and secure way.
- **Database Service:** this service can handle different database connections of different technologies for seamless integration.
- **Dataset service:** This can handle different data output and convert it into a relevant format or dataset for the user.
- **Webhook service:** a lightweight event-driven communication that automatically sends data between applications via HTTP.
- **Third-Party API Service:** this service will handle any third-party API service required for the task.
- **Other Services:** API Gateway allows the integration of any required services as per requirement.

The following tools are being developed in C2IMPRESS project that will be integrated and interconnected (Figure 2.3 and Figure 2.4) in the platform:

1. Database and Knowledge Graph
 - Ontology-Based Knowledge Graph
 - Centralized Database
2. Risk Management and Decision Support Tools
 - Multi criteria Analysis (MCA)
 - Spatial multi criteria decision analysis (SMCDA)
 - Horizon Action Eye
 - Big Data-Powered Decision Support Tool
 - DPSIR and PESTLE
3. Early Warning System
 - HIDRALERTA
 - Fluvial Flooding and Landslide in the Ordu, Turkey
 - Flood Hazards in the Mallorca, Balearic Island
 - Wildfire Hazards in the Balearic Island
4. Simulation Models
 - SoS4MHRIN
 - ABM & HBM
5. Risk & Resilience
 - Disaster Monitoring and Observation Portal (DMOP)
 - Risk & Resilience Framework
6. Guidelines & Methods



- Public- private-Civic partnership (PPCP)
- Citizen Engagement Framework
- ICCA-MHDMF
- Polycentric Risk Governance Framework
- Policy Roadmap





Figure 2.3: DSS Dashboard UI – Common User Interface

2.4.1 API Gateway

An API gateway is a system that acts as a single-entry point for all client requests made from the CUI to multiple backend services, typically in a microservices architecture. It simplifies and manages the



communication between clients (like web or mobile applications) and the various microservices that make up an application.

How API gateway connects Microservices:

- **Request Handling:** The API gateway receives a client request (e.g. a request to access MCA service).
- **Routing:** Based on the request type, the API gateway forwards it to the appropriate microservice(s).
- **Aggregation:** In some cases, the API gateway can aggregate data from multiple microservices and return a unified response.
- **Transformation:** It can modify requests and responses, handling format transformations, authentication or validation.

Benefits of API gateway

- **Load Balancing:** The gateway can distribute client requests across multiple instances of microservices, helping to manage load efficiently.
- **Security:** It provides a central place for handling security concerns like authentication, authorization, and SSL termination, which makes the architecture more secure.
- **Rate Limiting & Throttling:** This helps manage and control traffic by enforcing rate limits to avoid overload on microservices.
- **Versioning & Flexibility:** Supports API versioning and allows easier updates to individual microservices without affecting clients.
- **Protocol Translation:** It can translate between protocols (e.g., from HTTP to WebSocket), making it easier for microservices to evolve independently.

2.4.2 Embedded Tools

Embedding different tools or services to the CUI that already have a UI (user interface) is another way to integrate different independent services that have already developed their own authentication system. Duplicating the tool's UI to the CUI is redundant; instead, it's possible to take the key dataset from the service owner and embed it like a widget in the CUI.

Embedding refers to the process of inserting content from one source (such as a video, map, dataset, or image) directly into another platform or webpage so it can be viewed and interacted with without leaving the current page. The embedded content is hosted externally but displayed within the context of the website or application (in our case, it is the CUI), often through an HTML `<iframe>` or similar methods.

2.4.3 How Embedding is Done

Embedding typically involves using an HTML `<iframe>` element or other specialized tags. Here's a breakdown of the most common approaches:

1. **Embedding via `<iframe>`:** The most common method, an `iframe` acts as a window on a webpage that shows content from another source. Example:
2. **Embedding via Specific Embedding Tools:** Special Embedding code provided by the platform or service to display specific data. Example:

```
<iframe
  src="https://www.youtube.com/embed/VIDEO_ID"
  width="560"
  height="315"
  frameborder="0"
  allowfullscreen>
</iframe>
```


3. Using Embed

Widgets: Web services can provide built-in embed widgets, which can be

configured for how the embedded content will appear and generate an embed code to copy and paste.

```
<blockquote class="twitter-tweet">
  <a href="https://twitter.com/user/status/123456789"></a>
</blockquote>
```

2.5 Integration with C2IMPRESS Centralised Database

C2IMPRESS Centralised Database [T4.1] is designed with two interconnected databases: a document-oriented (NoSQL) database and a relational (SQL) database. The NoSQL database manages user authentication, file history, metadata, etc., while the SQL database stores processed geographic information system (GIS) data derived from uploaded files.

Key aspects of the design include seamless integration with an Ontology-Based Virtual Knowledge Graph (VKG) [T4.2] for interoperable data sharing and the ability to interface with other microservices developed by the C2IMPRESS consortium partners. Each uploaded file is linked to specific users through metadata stored in the NoSQL database, and the relational database catalogues all files for easy access. Detail description of the database is given in D4.1. To facilitate the reader of the D4.2, a very brief information is provided below.

2.5.1 Data Type

The Database System supports dynamic and flexible data types based on the files uploaded by case study area (CSA) partners for seamless integration with the Virtual Knowledge Graph (VKG). Different GIS file formats, such as shapefiles, TIFFs, and NetCDF, are accepted, and each file type triggers the creation of corresponding tables in the relational database.

The rest of the other data will be stored in the NoSQL Database. Taking this into account, there will be separate schemas for separate data collections. For example, in the T4.6 HazardMonitor App, the C2IMPRESS Users can upload images, provide coordinates about any local disaster scenario, etc. Those data will be stored in the NoSQL database according to their corresponding schemas. A detailed description of the schemas is provided in every corresponding microservices section (Chapter 3).

2.5.2 DSS Architecture & Database Integration

The architecture of the C2IMPRESS project's centralised database is designed to support a robust and flexible data pipeline that facilitates cataloguing, transforming, and storing various datasets (Figure 2.4). The system integrates seamlessly with an ontology-based Knowledge Graph (KG) to ensure data interoperability, a key feature of the project. Additionally, the database is designed to interact with microservices, open-source tools, and software, such as the Disaster Monitoring and Observation Portal (DMOP), Hazard Monitor App, Virtual Knowledge Graph Dashboard, Multi-Criteria Analysis (MCA) Tool, DPSIR & PESTLE Tool, etc.

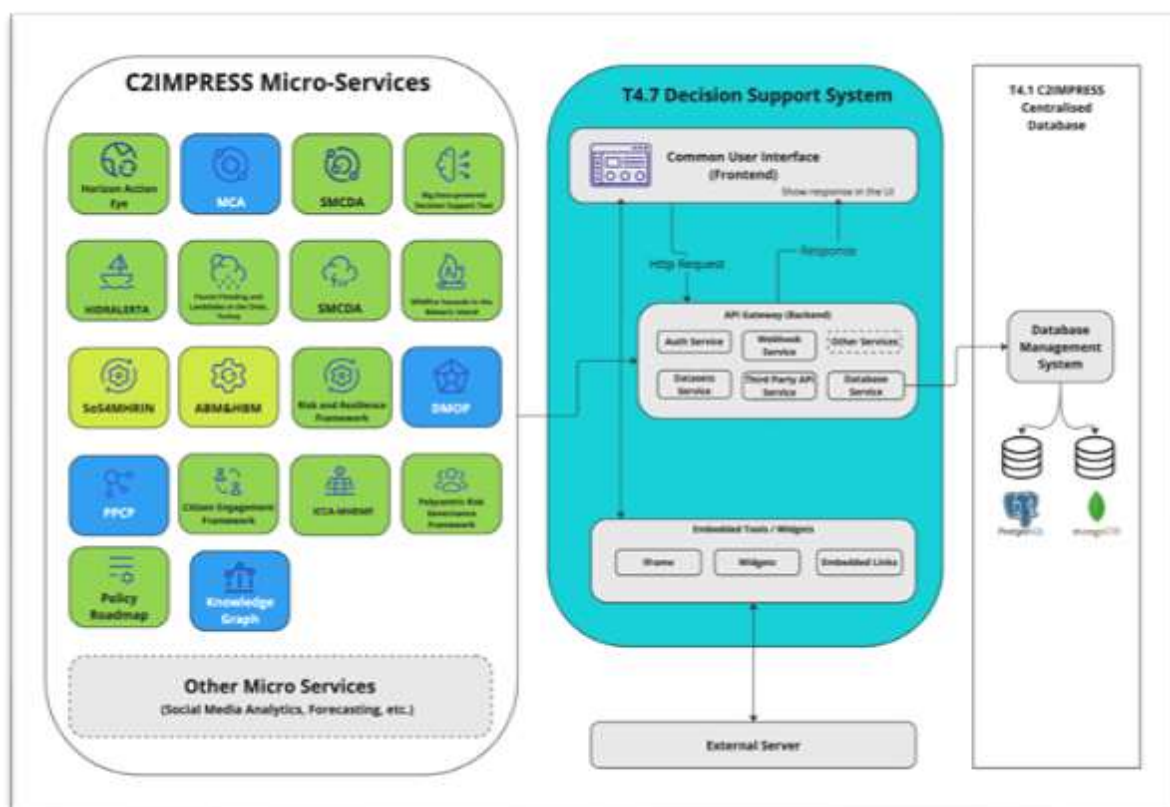


Figure 2.4: DSS Architecture & Database Integration

2.5.3 Integration with Ontology-Based Virtual Knowledge Graph

The integration with VKG part has three primary components:

1. **Data Organisers:** This component allows users to upload files to the platform. Users can add metadata such as region, year, and country, which helps organize the data into a catalogue. A user interface is provided for partners to upload, track, and download files.
2. **Data Catalogue:** Once the files are organized, they are stored in the Data Catalogue. The catalogue stores the sequence of uploaded files and their associated metadata, facilitating easy access and retrieval.
3. **Data Transformation:** This component processes the uploaded files, transforming the raw data into structured formats. Supported file types include Shapefile, Raster, NetCDF, GRIB, CSV, and GEOJSON. Each transformed file generates a new table in the relational database, allowing for structured storage and future querying.

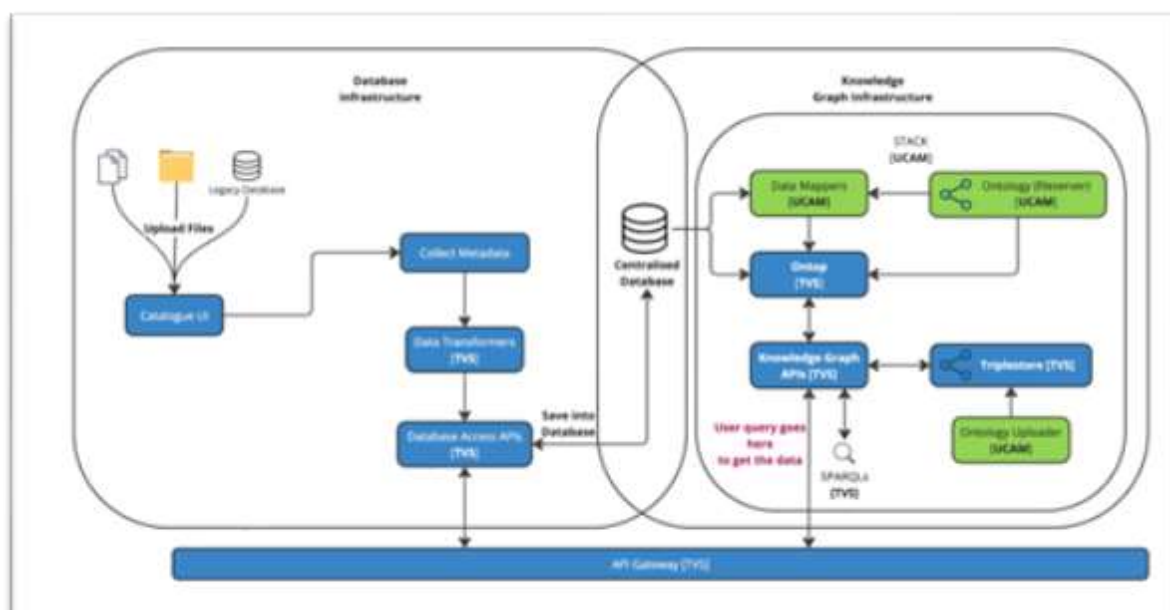


Figure 2.5: Integration with DSS Centralised Database and Ontology-Based Virtual Knowledge Graph

The architecture also features a user-friendly interface that enables partners to manage their accounts, upload files, monitor processing progress, and view data transformation statuses. This ensures an efficient and transparent workflow from data upload to transformation and cataloguing, making it easier for partners to access and utilize the standardized data.

The detailed demonstration of this architecture is described in D4.1. The user interface for VKG is in progress.

2.5.4 Integration with DMOP & Hazard Monitor App

The Disaster Monitoring and Observation Portal (DMOP), which is T4.6, is also seamlessly integrated with the Centralised Database. DMOP operates as a microservice within the DSS platform, leveraging the database designed in Task 4.1. The DSS platform will feature an API Gateway to manage internal and third-party API services, including modules that depend on external data sources and webhooks for real-time data exchange.

Several key modules within DMOP rely on third-party services:

1. **Forecasting:** The platform will use OpenWeather API [12] to retrieve real-time weather data for disaster forecasting.
2. **Disaster Impact Analysis:** Data on recent disasters and their impacts.
3. **Social Media Awareness:** This module collects disaster-related activity from social media platforms, planned to integrate with the DSS platform with an iframe, which will be provided by SAMPAS (T4.3).
4. **Early Warning System (EWS):** Like the Social Media Awareness module, the EWS will receive data through C2IMPRESS Consortiums.

Figure 2.6 shows three interfaces of the DMOP that have been already developed under T4.7

A detailed description of DMOP can be found in Deliverable D4.1.


Home Dashboard

Ornop Dashboard

- Weather Monitoring
- Recent Hazards
- Early Warning System
- Forecasting
- Risk Assessment
- Disaster Losses and Damage Tracking
- Multi-Hazard Disaster Impact**
- Legacy Monitoring Service
- Climate

Multi Hazard Disaster Impact Analysis

The module shows annual and decadal reports of the casualties caused by different disasters across Greece, Turkey, Spain and Portugal

Source: <https://ppluworlddata.org>

Select Country
Turkey

Select Report Type
Annual Decadal

Disaster Impact Analysis

Year	Country	Disaster	Impact Details
2023	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2014	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2015	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2017	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2018	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2019	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2020	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2021	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2022	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact
2023	Turkey	Droughts, Earthquake, Flood, Storm, Wildfire, Extreme Temperature...	See full impact

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Home Dashboard

Ornop Dashboard

- Weather Monitoring
- Recent Hazards**
- Early Warning System
- Forecasting
- Risk Assessment
- Disaster Losses and Damage Tracking
- Multi-Hazard Disaster Impact
- Legacy Monitoring Service
- Climate

Recent Hazards - Worldwide

The module provides recent hazard details from all over the world

Source: <https://www.usda.gov>

All Hazards

 Flood

 Earthquake

 Cyclone

 Wildfire

 Volcano

 Drought

Flood

Page Size: 10

Title	Occuring Time	Affected Areas	Alert Level	Details
Flood in Romania...	0 Oct 2024	Bosnia & Herzegovina	Orange	See details
Flood in Nepal	28 Sept 2024	Nepal	Orange	See details
Flood in Austria, Cze...	18 Sept 2024	Austria, Czech Republic, Germany, Poland, Romania, Slovakia	Orange	See details
Flood in India	16 Sept 2024	India	Orange	See details
Flood in Sudan	6 Aug 2024	Sudan	Orange	See details
Flood in Indonesia	02 Jun 2024	Indonesia	Orange	See details
Flood in Afghanistan	25 May 2024	Afghanistan	Orange	See details
Flood in Brazil	17 May 2024	Brazil	Orange	See details
Flood in Kenya	4 May 2024	Kenya	Orange	See details
Flood in Burkina F...	3 May 2024	Burkina Faso	Orange	See details

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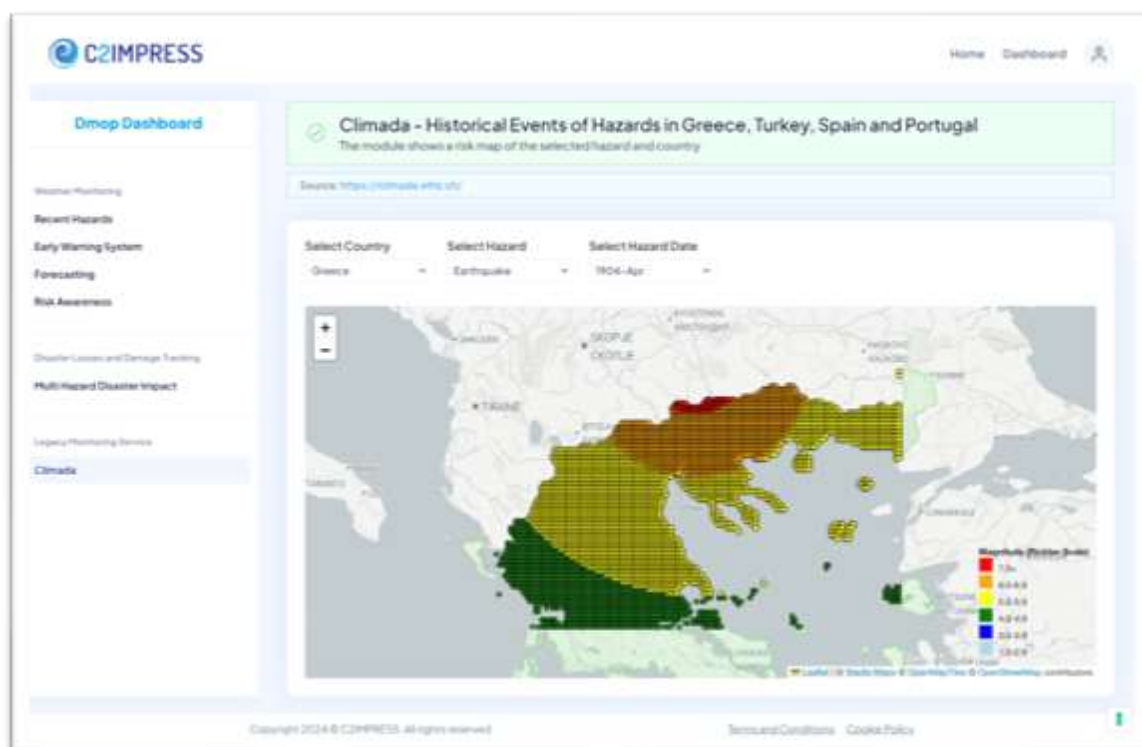


Figure 2.6: Three examples of DMOP User Interfaces

Hazard Monitor App:

The **Hazard Monitor App**, which is T5.2, is also developed and connected with The Centralised Database. The **Hazard Monitor App** is consisting of three primary modules –

- **Recent Hazards:** Citizens can view nearby active disasters around the world filtering by several disasters. By clicking on the hazard name, citizens can access more detailed information such as origin, affected area, and coverage, danger level, etc. These active hazards information is integrated and coming from [GDACS \[14\]](#).
- **Disaster and Impact Observation:** This module allows citizens to identify more information about the magnitude and impact of previous disasters in the area. This feature also aims to increase citizens' awareness by providing updates on the impacts of relevant disasters for our CSA Countries. The impact information covers wide range of data between 1900 to 2023 years. Disasters Impact Information is pulled of [OurWorldInData \[13\]](#).
- **Citizens Forum:** Citizens Forum is designed with two primary sections –
 - **Emergency Services:** This module is focused to provide easiest way to contact local emergency services in any emergency. In the **Error! Reference source not found.**, all the emergency services are listed.
 - **Awareness & Helpful Tips:** In Citizens Forum module citizens can upload image, of the affected area. They can see their own uploads as well as images uploaded by others. Citizens can also provide description of the disaster while uploading so that in future others can know the disaster types of that specific area. A machine learning model will be used to process this image uploaded by the citizens. This model will determine the water level by using Object Identifying Model. This model requires images should contain a person standing over the water. By using the reference point of the person's height, this model can determine the water level more accurately. If any image does not contain a person, the model returns an error.

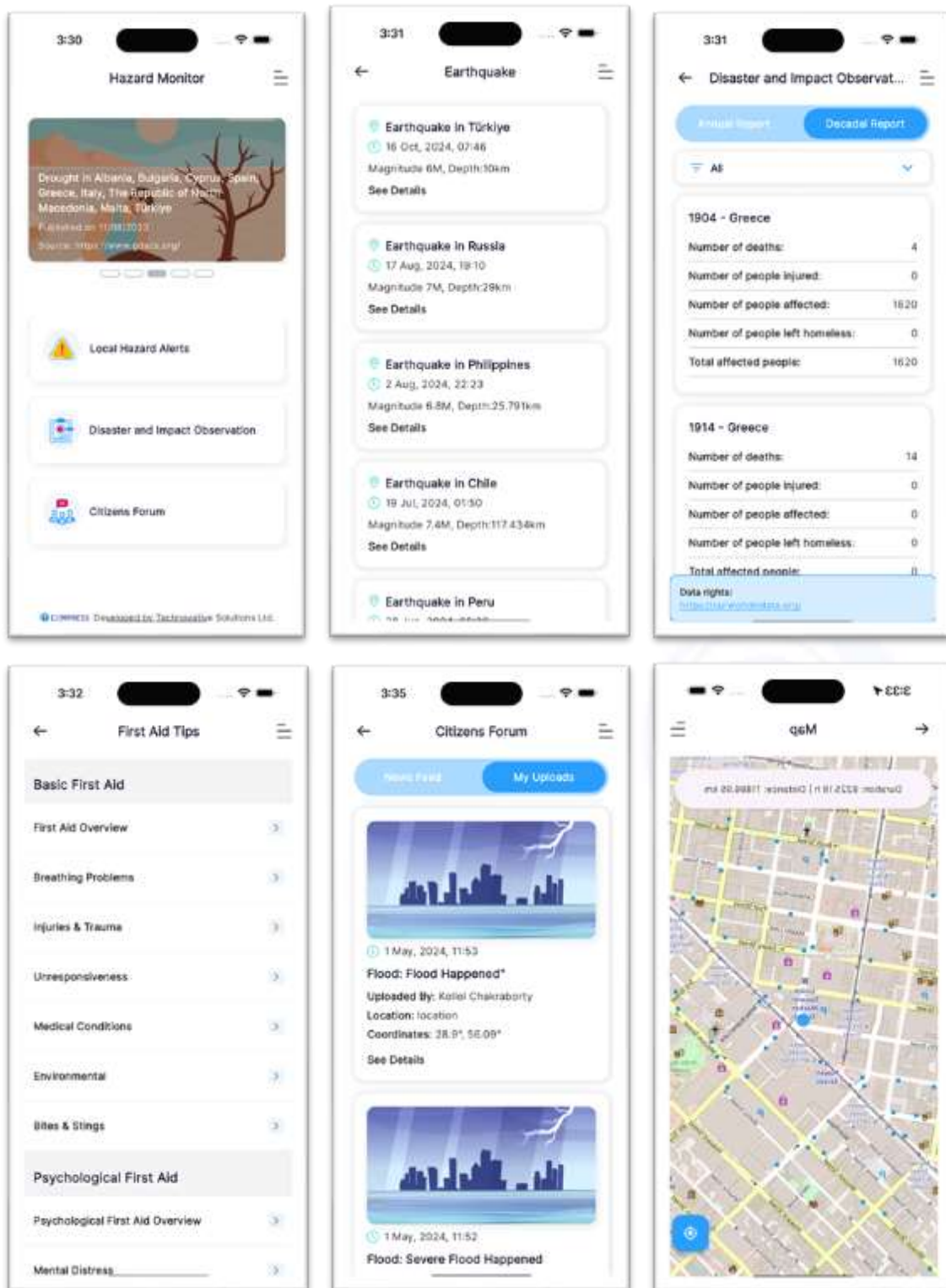


Figure 2.7: Hazard Monitor App User Interface

2.5.5 Integration with MCA Tool

The Multi-Criteria Analysis (MCA) Tool is an effective decision-making tool because it systematically analyses and assesses several possibilities or alternatives against a set of predetermined criteria. The MCA tool is integrated into the DSS Dashboard(D4.2).

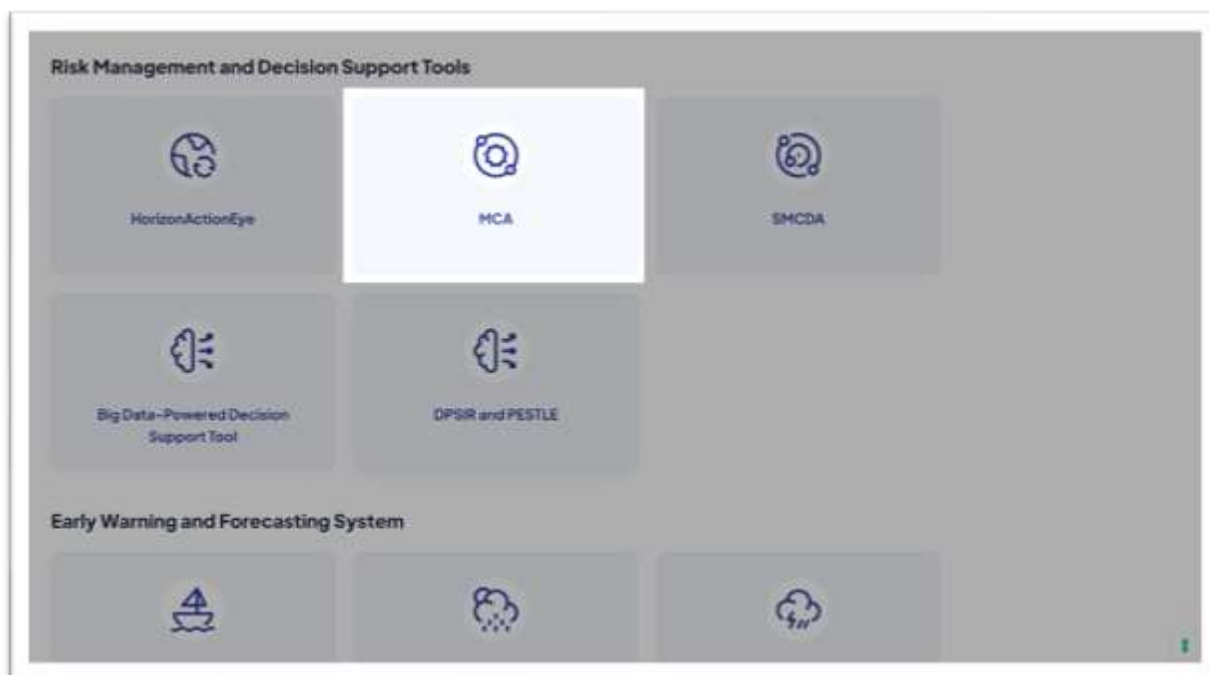
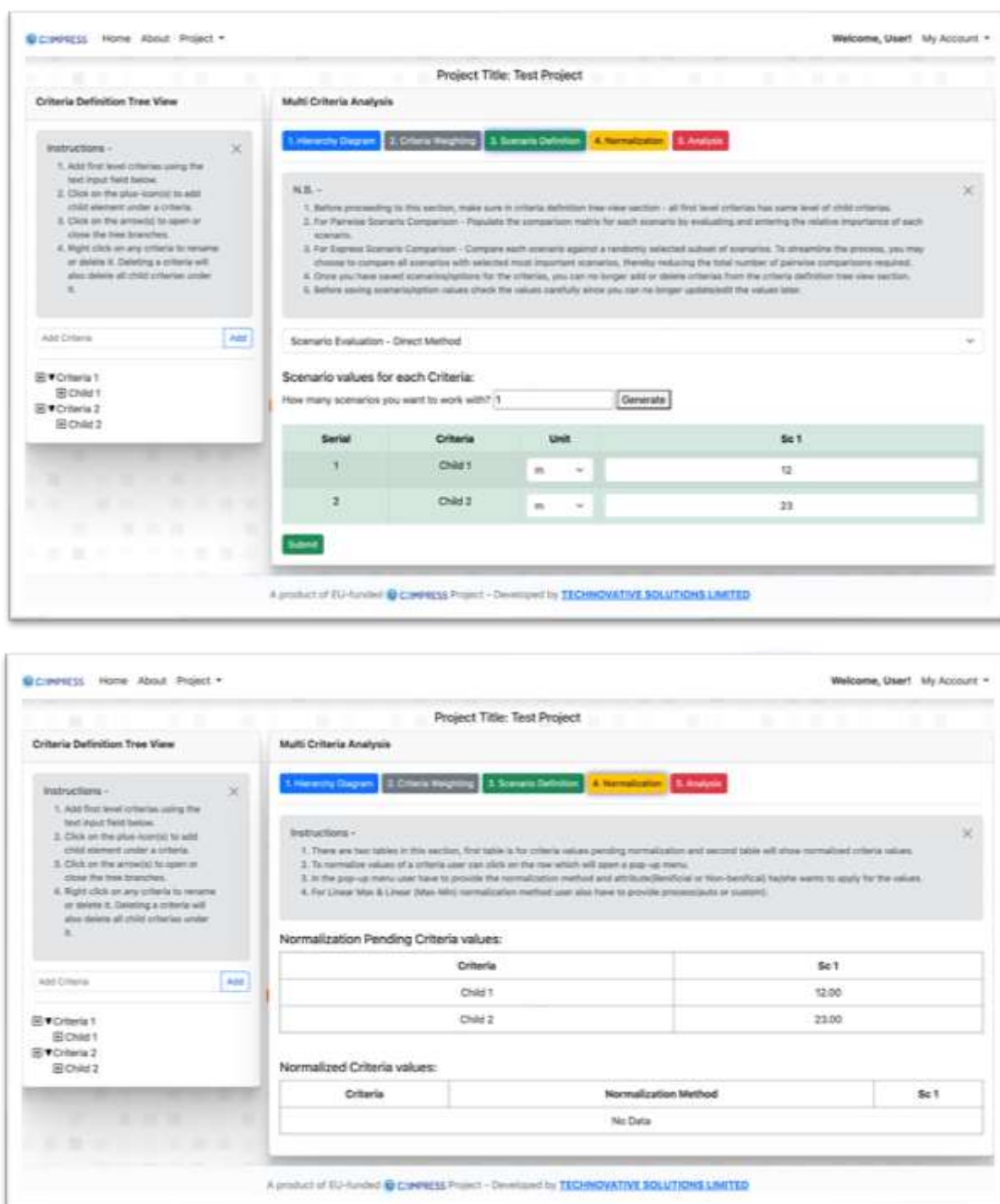


Figure 2.8: MCA Tool from DSS Dashboard



Project Title: Test Project

Multi Criteria Analysis

1. Hierarchy Diagram 2. Criteria Weighting 3. Scenarios Definition 4. Normalization 5. Analysis

Instructions -

1. Add first level criteria using the text input field below.
2. Click on the plus icon(s) to add child element under a criteria.
3. Click on the arrow(s) to open or close the tree branches.
4. Right click on any criteria to rename or delete it. Deleting a criteria will also delete all child criteria under it.

Add Criteria

▼ Criteria 1
 Child 1
 ▼ Criteria 2
 Child 2

Scenario Evaluation - Direct Method

Scenario values for each Criteria:

How many scenarios you want to work with?

Serial	Criteria	Unit	Sc 1
1	Child 1	m	12
2	Child 2	m	23

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Project Title: Test Project

Multi Criteria Analysis

1. Hierarchy Diagram 2. Criteria Weighting 3. Scenarios Definition 4. Normalization 5. Analysis

Instructions -

1. There are two tables in this section, first table is for criteria values pending normalization and second table will show normalized criteria values.
2. To normalize values of a criteria user can click on the row which will open a pop-up menu.
3. In the pop-up menu user have to provide the normalization method and attribute (Beneficial or Non-beneficial) he/she wants to apply for the values.
4. For Linear Max & Linear (Max-Min) normalization method user also have to provide process (auto or manual).

Normalization Pending Criteria values:

Criteria	Sc 1
Child 1	12.00
Child 2	23.00

Normalized Criteria values:

Criteria	Normalization Method	Sc 1
	No Data	

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Figure 2.9: MCA Tool User Interface

2.5.6 Integrated DPSIR and PESTLE tool

Driver, Pressure, State, Impact, Response (DPSIR) & Political, Economic, Social, Technological, Legal, and Environmental (PESTLE) Tools are developed as a separate and independent module. It is related to the Decision Support System (DSS) platform by redirecting to the tool from the DSS dashboard. By clicking on the link from the dashboard (D1.1 and D4.2).



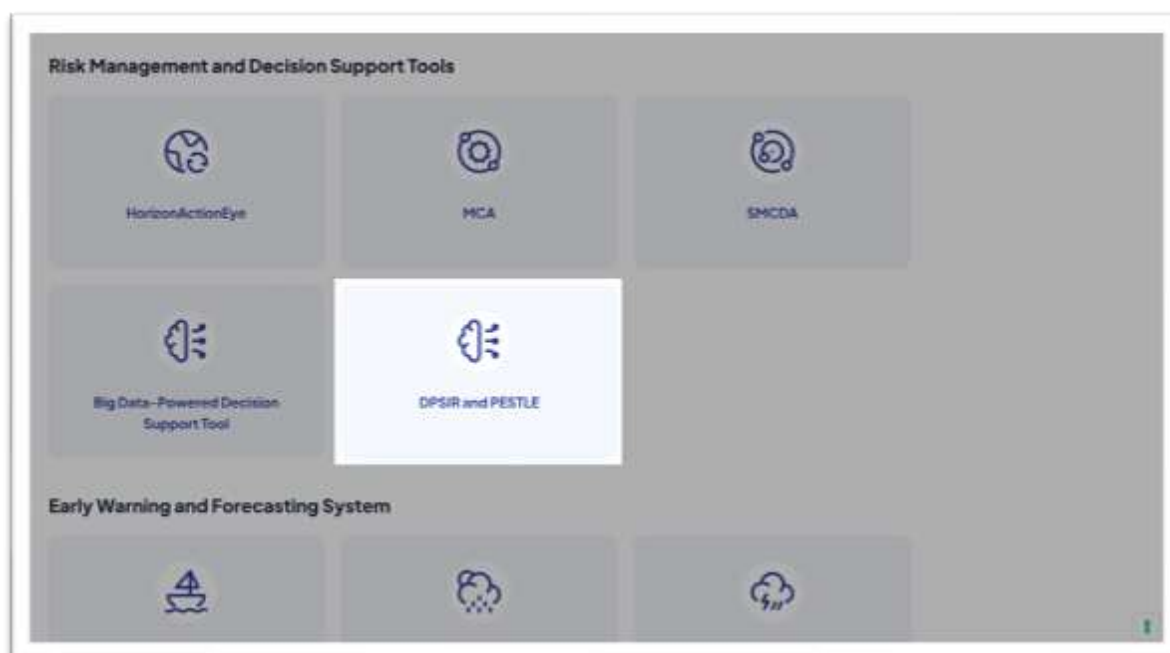
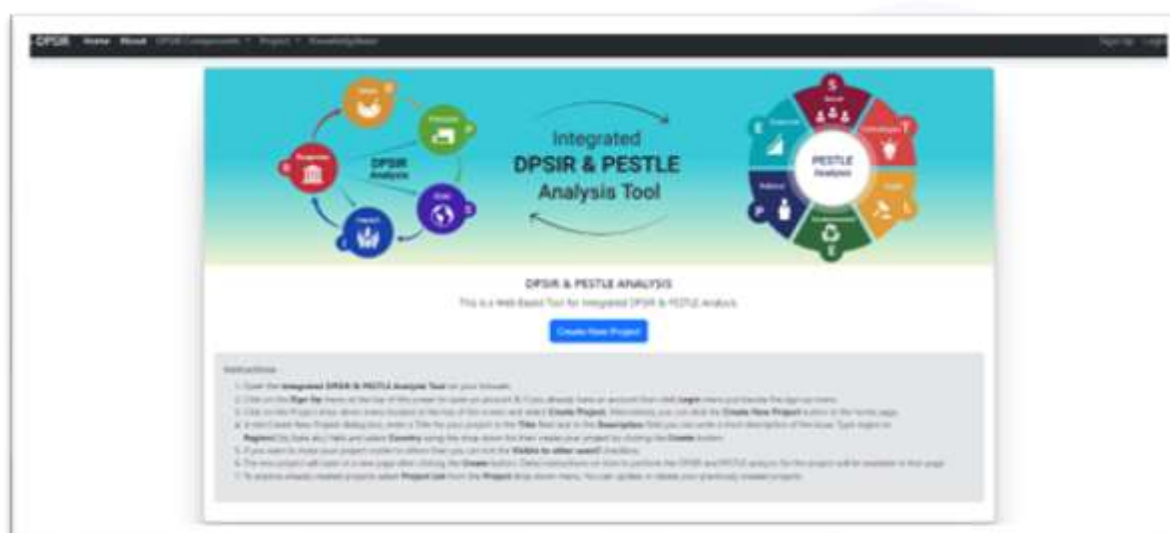


Figure 2.10: DPSIR & PESTLE Tool from DSS Dashboard



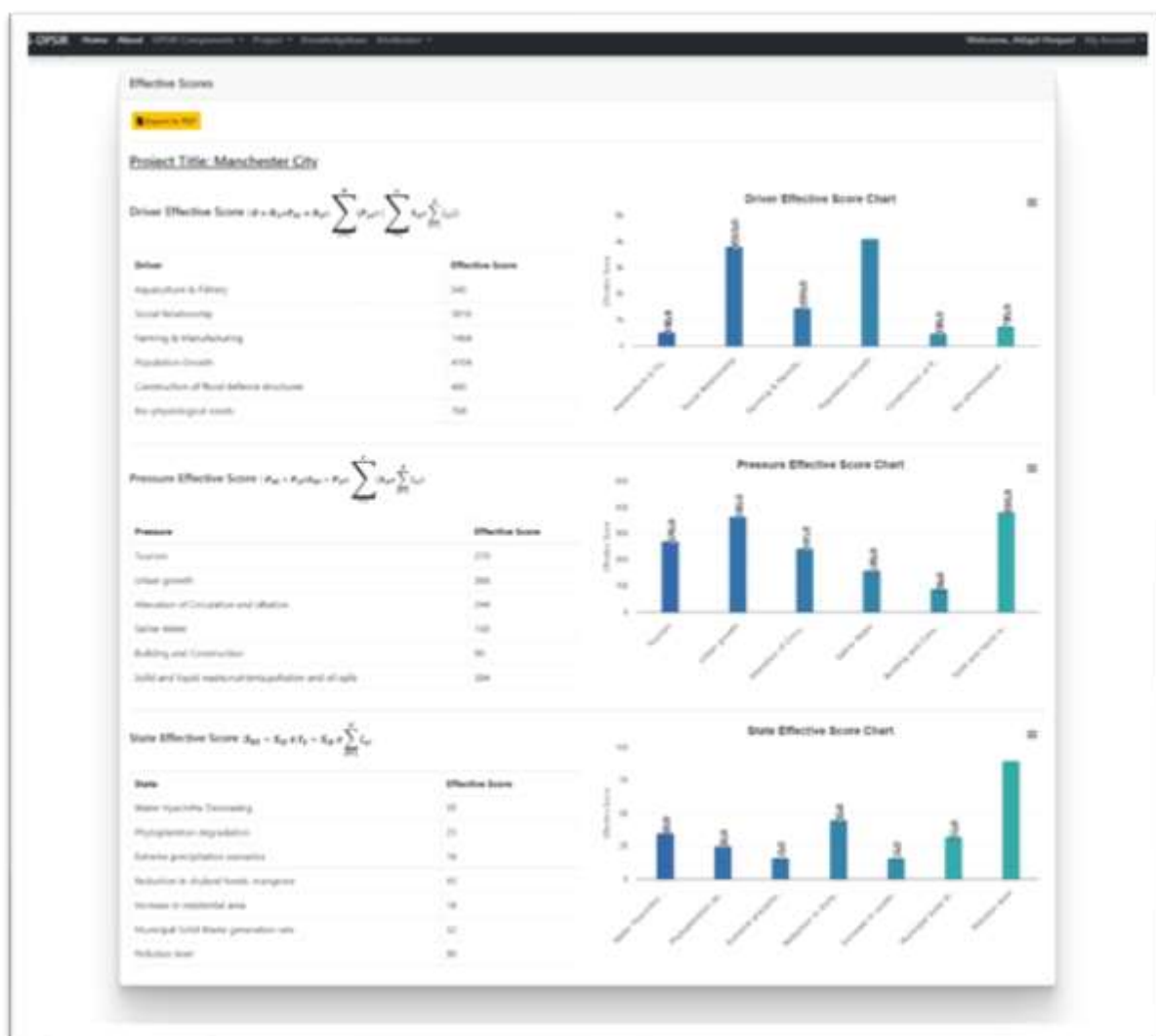


Figure 2.11: DPSIR & PESLTE Tool User Interface

3 Brief Summary of C2IMPRESS Microservices

3.1 System-of-Systems for Multi-hazard Risk Intelligent Network (SoS4MHRIN)

The System-of-Systems multi-hazard risk Intelligence Network (SoS4MHRIN) framework developed at the C2IMPRESS project provides dynamic analytics to assess the complete picture of the multi-hazard risk cycle in a multi-scale, multi-domain manner. For that purpose, SoS4MHRIN provides new dynamical system analytics and a model design framework targeted to Multi-Hazard Intelligence articulation grounded on MET's IPAI-ESDI (Information Physical Intelligence and Earth System Dynamic Intelligence) on a synergistic interplay between information physics, Artificial Intelligence (AI) and Earth system dynamics. This innovative tool further underlies the novel end-user products and their benchmarking relative to State-of-the-Art (SoA) models so that the users can consult, analyse and visualise both SoA and novel outputs such as those about hydrometeorologic and hydroclimatic quantities. It is thus being developed for assessing, processing, simulating, analysing, forecasting, and evaluating causal processes and interactions, from inputs to process impacts, and demonstrating the entire modelling capacity to provide the complete picture of the climatic cycle and climate system dynamics, and the articulated socio-environmental coevolutionary interplay, along with the overall system dynamic framing of interacting multi-hazards. The platform is further integrating anthropogenic processes and interactions such as through human agent-based behavioural models about the intrinsic dynamics between humans and the multi-hazard risk structure, e.g. in terms of



awareness and response, to raise awareness further, improve understanding and strengthen human preparedness and response at the onset of critical hazardous and multi-hazardous events. This incorporates cognitive reasoning based on human behaviour models and includes a novel traffic model to capture the mobility of the affected population and a decision model to inject a proposed hazard response by the authorities.

3.2 HIDRALERTA: Forecasting and Early Warning System for Coastal Flooding and Ship Berthing

The present version of HIDRALERTA system uses a combination of numerical models to simulate wave propagation, estimate mean overtopping discharge over port infrastructures and coastal defence structures, and determine ships' motions and mooring forces. During C2IMPRESS, HIDRALERTA will be further extended to consider the influence of fluvial flooding on ships' motions and mooring forces, and ship maneuvering. HIDRALERTA contributes to the C2IMPRESS project's aim of better understanding the actual and climate change impacts associated with coastal and fluvial floodings, ship manoeuvring and mooring, using past data and climate change scenarios. This is possible because HIDRALERTA can be used as an EWS or a modelling system forced with pre-defined scenarios. It issues alerts for overtopping and induced flooding in coastal areas and harbours and for moored ships inside the port or harbour. Moreover, the system may also act as a long-term management tool since it can simulate the response to future scenarios related to climate change, such as the increase of the mean sea level and storm severity, which will increase coastal flooding probability. This system is divided into four main modules. Module I is for sea state characterisation, where numerical models propagate offshore wave conditions (hindcast or forecast data from the European Center for Medium Range Weather Forecast (ECMWF) and the European Union's Copernicus Marine Service (CMEMS)) to nearshore and inside ports. Module IIa determines Wave Run-up and Overtopping, using the sea conditions obtained in Module I. Module IIb is for Ship behaviour, where wave effects in terms of UKC of maneuvering ships and forces on mooring lines and fenders and the motions of a ship moored at a quay are determined using numerical models. Risk Assessment is analysed in Module III by comparing the relevant computed values (overtopping discharge, maximum run-up, ship movements and mooring forces) with pre-set thresholds. Module IV is for the Warning System, which includes a web platform, 72-hour forecasts (with 3-hour intervals) updated daily, and all the results from the previous modules (including wave conditions, wave overtopping discharges, ship motions and forces on the mooring system).

3.3 Forecasting and Early Warning System for Fluvial Flooding and Landslides in Ordu, Turkey CSA

The Forecasting and Early Warning System (EWS) for Fluvial Flooding and Landslides in Ordu, Turkey, developed by ORDU within the C2IMPRESS project, represents a crucial advancement in disaster resilience. Despite previous hydrodynamic model results suggesting that the region may be less susceptible to floods based on 500-year frequency river flow calculations, recent climate events underscore the necessity for revised criteria. Focused on the Civil River in Altınordu İlçesi, this system aims to mitigate the risks of fluvial flooding and landslides, especially in urban areas with dense infrastructure. Utilising LIDAR-supported drone surveys, EWS stations in three strategic points along the Civil River have been installed. These stations capture real-time data on instantaneous flow, depth, and visual observations, contributing to a comprehensive understanding of flood dynamics and landslide potential. The data collected by these stations serve as the foundation for updated flood modeling, which is vital for adapting to climate change scenarios and revising flood recurrence flow rate calculations. The EWS provide invaluable insights into time-dependent water level changes, precipitation-flow rate relationships, and topographical influences, facilitating more accurate and responsive disaster management strategies. Integrating these data into flood modeling studies promises to enhance preparedness and response efforts, enabling stakeholders to make informed decisions in safeguarding lives and infrastructure against the growing threat of multi-hazard events. In the broader context, the ORDU-developed EWS exemplifies the objectives of the C2IMPRESS project, which seeks to shift disaster resilience frameworks from hazard-centric approaches to holistic,



people-centred strategies. By leveraging technology and interdisciplinary collaboration, the Forecasting and Early Warning System for Fluvial Flooding and Landslides in Ordu, Turkey, pave the way for more resilient societies facing escalating climate risks.

3.4 Forecasting and Early Warning System for Flood Hazards in Mallorca, Balearic Island CSA

The University of the Balearic Islands, via the Natural Hazards and Emergencies Observatory of the Balearic Islands (RiscBal), in collaboration with the Government of the Balearic Islands (GOIB), implemented an Early Warning System for floods in the Balearic Islands (INUNSA). This (and further) EWS is based on three pillars: RiscBal-Control, RiscBal-Data and RiscBal-Alerts. RiscBal-Control is an environmental monitoring infrastructure with 99 self-owned stations that deliver real-time data on river conditions (water level and discharge), soil moisture levels, and precipitation every 5-10 minutes. RiscBal-Data defines a data model for the hydrometeorological network, aiming to improve data efficiency (type and structure) and facilitate continuous and automatic monitoring of river flow and hydrometeorological parameters. It is based on different PostgreSQL-PostGIS databases, where real-time and semi-static data is stored to perform all the analysis on the other servers, and results are provided through an online viewer. RiscBal-Alerts is the proper EWS for flood risk, integrating RiscBal-Control and RiscBal-Data, as well as simulation and modelling tools. Three risk levels are defined to identify a flood's potential impacts and losses: Level R1) rural areas; Level R2) Peri-urban/rural areas; 1st and 2nd order roads and highways; and Level R3) urban areas and all types of roads. Warning levels were defined based on the water level, in percentage, and the risk levels previously mentioned; therefore, a combination of the risk level of the station's surroundings and the water level is used to spread an alert to the affected station and extended to downstream stations. This prompt alert at the headwaters of the catchment would allow emergency services to prepare the vulnerable areas downstream for the flood. In addition, RiscBal has developed a tool to obtain the runoff threshold using a curve number to identify specific locations within a territory where rainfall at a given moment could turn into surface runoff, resulting in a concentration of flow towards watercourses and potentially leading to adverse impacts. Utilising this data, 24–48-hour flood risk prediction reports are generated. The utilisation of these tools facilitates the establishment of a Flood Warning and Emergency Management Service.

3.5 Forecasting and Early Warning System for Wildfire Hazards in the Balearic Islands

The University of the Balearic Islands, via the Natural Hazards and Emergencies Observatory of the Balearic Islands (RiscBal), in collaboration with the Government of the Balearic Islands (GOIB), aims to develop an EWS for wildfire hazards based on the RiscBal infrastructure aforementioned: RiscBal-Control, RiscBal-Data, and RiscBal-Alerts (see more details at <https://riscbal.uib.eu/>). The core of this microservice is the same as forecasting and early warning systems for flood hazards in Mallorca, Balearic Island CSA. Data from meteorological stations are the basis for a set of indicators that were used to assess the wildfire risk in the Balearic Islands. In addition, optronic cameras were implemented to better monitor terrestrial radiation. All these real-time data were complemented with remote sensing data on soil moisture, vegetation density, health, or other useful indicators.

3.6 FlowBasin – web-based app for watershed characterization and flow evaluation

LNEC is developing an innovative web-based platform for C2IMPRESS that integrates topographic data from the European Digital Elevation Model (EU-DEM) and utilises simple hydrological models, leveraging Pysheds, an open-source library, to delineate and characterize watersheds. The tool allows users to choose a region of interest and execute several analytical procedures. Upon selecting an area of interest, Pysheds computes the flow direction from the EU-DEM, delineates catchments based on the flow direction, and calculates flow accumulation. The river network is then extracted at a specified accumulation threshold, prompting users to select a point within the network. Ultimately, the watershed is delineated and essential characteristics such as area, perimeter, hypsometric plot, and



concentration time are provided. The platform's user-friendly interface may help users, including water resource managers and decision-makers, to comprehensively visualise watershed characteristics and flow discharge patterns. This capability significantly enhances the efficiency of planning decision-making processes in water resources, facilitating rapid and well-informed decision-making. The FlowBasin platform will improve its capabilities to provide more comprehensive water resource management insights, incorporating the ability to estimate inundation areas (using HAND methodology). Additionally, we aim to integrate features for characterising precipitation patterns, allowing users to understand better rainfall's spatial and temporal distribution within watersheds. Furthermore, considering the increasing impact of wildfires on hydrological processes, future iterations of the platform will include modules to assess the influence of wildfires on flow discharge dynamics, providing valuable insights for managing water resources in fire-prone regions. These enhancements will further enhance the FlowBasin platform as a tool for informed decision-making in water resources planning and management.

3.7 HazardActionEye: Decision Support Tool for Response Planning During Crisis

HazardActionEye is an automated and interactive decision support tool that generates infrastructure protection plans and an immediate response strategy in the form of a decision tree, with maximum possible coverage in the light of time and resource limitations. The preparatory developments took place within the scope of SoS4MHRIN, which provided input for the HazardActionEye. This decision support tool is set to improve the multiple parallel simulation algorithms and novel components to monitor simulation runs, terminate them, and start new runs originating from existing ones according to domain-specific decision scenarios. To efficiently monitor ongoing simulations, the tool complemented the automatic processes with user involvement via interactive visualisations, such as geospatial visualisations of dynamic changes in the ongoing simulations, attracting the user's attention to interact with ongoing simulations and steer them. Additionally, the tool produced visual signs enabling the user to adjust the desired simulation properties for important regions or infrastructures, leading the simulation process to adjust accordingly. The tool also entailed a complex decision tree algorithm, tested and validated with site-specific data and provided visualisation and integration capabilities into the overall C2IMPRESS platform. The pertinence of this tool to C2IMPRESS and beyond stemmed from the increased mathematical soundness, physical consistency, and operational capabilities, both passive in terms of providing simulations and their associated visualisation and active in terms of allowing for actionable user interaction with the simulation process to assist in the live operational response, be it in preparatory evaluations and exercises to train for emergencies, during the actual occurrence of such with real-world information at play, or afterwards to look back for insights and lessons to be learned towards further improvement of methods and practices in effectively handling future events.

3.8 Integrated Multi-hazard Disaster Risk and Resilience Assessment Framework

The Integrated Multi-Hazard Disaster Risk and Resilience Assessment Framework is a multi-actor framework developed to develop a novel integrated multi-hazard disaster impact and risk assessment framework and a novel community resilience framework considering risk prioritisation. These frameworks aim to be used as decision-support tools, permitting the research team and the representatives of the Case Study Areas to perform a multidimensional vulnerability assessment concerning physical, social, institutional, economic, and environmental susceptibility to the most prominent hazards of their community, drive risk-informed decision-making, and assess their changing capacities for societal resilience. The C2IMPRESS risk and resilience microservices are two interconnected services based on a mix of user-provided information, data from modeling and simulations, especially concerning climate, single and multi-hazard events, and feeds from global, national, and local databases. For the risk microservices, the user defines the communities, services, ecosystems at risk, & acceptable levels of risk along with the categories for the hazard, exposure, impact, and risk assessments, including impact/vulnerability models and curves. For the resilience



microservices, the user defines capacities/indicator values for the resilience microservices for the selected categories. The Disaster Risk and Resilience Assessment Framework consolidates the general aim of the C2IMPRESS project by collecting and providing both quantitative and qualitative data concerning natural hazards and environmental characteristics, as well as data regarding the communities' social and cultural aspects. In addition, the frameworks provide an in-depth and innovative model to better understand each implementation region's exposure and vulnerability levels by thoroughly measuring its risk and resilience capacities.

3.9 PPCP-based Polycentric Risk Governance Framework

Artelia developed a polycentric governance framework to improve natural disaster management. This polycentric governance framework should enable a more effective and flexible response that is better adapted to each emergency's specific needs, enabling power and responsibility to be distributed between several autonomous or semi-autonomous decision-making centres. A generic framework was produced, and detailed action plans were developed for each CSA. The polycentric governance framework and the detailed action plans developed based on the PPCP approach were designed, tested, and validated, leading to the implementation of a governance model by round-trip between CSAs. This PPCP approach mobilises innovative social intelligence techniques (Design Thinking, Living Labs) to bring together the public, private and civil stakeholders of each CSA in the co-creation and co-design of solutions adapted to their territory. In concrete terms, PPCP-Living Labs (LL) are organised at the start of the C2IMPRESS project and gather local stakeholders four times, each six months apart, to participate in co-designing tools with C2IMPRESS partners.

3.10 Interactive Spatial Multicriteria Decision Analysis (SMCDA) Tool for Vulnerability, Hazard, Exposure, and Risk Quantification and Prioritization

Individual hazard maps can be helpful during the early stages of land development and urban planning. Still, they cannot accurately represent the complex scenario of multi-hazards. Multi-hazard risk maps offer a more unified scenario of multiple natural hazards, making them a better option for planners to determine which places are most at risk, evaluate the effectiveness of different adaptation strategies, and create appropriate hazard risk mitigation strategies. A spatial multi-criteria analysis (SMCDA) tool in the C2IMPRESS project is developed to support the classification of risk according to the pertinent conditions of a region. The tool generates a risk map based on the selected influencing factors of each hazard based on previous studies and data availability. Some factors are common for every hazard type, while others are particularly relevant for certain hazard types. The risk and resilience framework can be considered when selecting the socioeconomic factors for a specific study area by evaluating the resilience score of the community's socioeconomic indicators. This helps to identify the indicators that are most vulnerable during a hazard. Once these factors are identified, these spatial data are converted into decision criteria maps. Then, this tool incorporates a machine learning algorithm with the Analytical Hierarchy method (AHP) for multi-hazard risk assessment and mapping. Machine learning generates the hazard map based on the selected influencing factors. In contrast, the Analytical Hierarchy method (AHP) produces a socio-economic vulnerability map integrating socioeconomic factors. The final hazard risk assessment map combines the hazard and vulnerability maps. Finally, the different risk maps are superimposed based on their weights to generate a multi-hazard risk map for a specific study area. The weights of each hazard will be determined through a pairwise comparison of the historical hazard events of a particular area of study. Using this tool, decision-makers can create efficient evacuation plans for impoverished areas by integrating socioeconomic data, such as housing and population data, with the vulnerability map of hazards. Efficient evacuation planning can significantly reduce the number of casualties and injuries. Additionally, a risk map can aid in communicating with locals who are affected by hazards.

3.11 Decision Support Tool with Integrated Human Behavior and Agent-based Model for Hazard Simulation and Action Planning

The importance of practical and realistic evacuation simulation cannot be overstated, particularly in the face of hazards that demand rapid and efficient response strategies. Within the C2IMPRESS project, focusing on case studies such as earthquakes in Egaleo, Greece, floods in Mallorca, Spain, and bushfires in Mallorca, Spain, the development of a Decision Support Tool that integrates Human Behavior Modeling (HBM) with Agent-Based Modeling (ABM) represents a significant advancement in simulating evacuation processes with a high degree of realism. This tailored approach enables emergency planners to account for the diversity of human actions specific to each hazard scenario, thereby significantly improving pre-disaster preparedness and in-disaster response capabilities. Traditional evacuation models often rely on simplified assumptions about human behaviour, which can lead to inaccuracies in predicting evacuation outcomes. By incorporating the Belief-Desire-Intention (BDI) model for human decision-making into the ABM simulation environment of MATSim, our tool is uniquely positioned to achieve a more precise depiction of individual priorities, concerns, and varied social behaviours, thus reflecting potential real-world observations and responses to emergency scenarios. As a result, by simulating different evacuation scenarios, this tool enables emergency planners, policymakers, and stakeholders to identify optimal evacuation policies, routes, and congestion points, to name only a few, for improved evacuation planning and execution. The tool supports creating more resilient communities by providing insights into how different factors, ranging from urban design and demographic profiles to evacuation policies and individual behavioural responses, affect evacuation dynamics. It aids in identifying vulnerabilities within current disaster response strategies and offers evidence-based recommendations for enhancing evacuation efficiency and safety. In conclusion, the Decision Support Tool with Integrated Human Behavior and Agent-based Model for Hazard Simulation and Action Planning exemplifies the profound impact of combining interdisciplinary research and practical application in the C2IMPRESS project. The tool significantly advances our capacity for effective disaster response planning by utilising advanced simulation technologies alongside in-depth analyses of human behaviour. This innovation not only improves immediate disaster response strategies but also furthers the broader mission of fostering safer, more resilient communities across the globe.

3.12 A tool for social media analytics to address multi-hazard risk management

Social media data plays a significant role in hazard research due to its real-time nature and vast reach. By analysing social media posts, researchers can swiftly identify and track the onset, progression, and aftermath of natural disasters or public health emergencies. The architecture of the C2IMPRESS Social media data analytics platform consists of different pillars that enable the gathering of data from various sources to analyse, store, and visualise the vast amount of information, both historical and in real-time, into a single platform. This enables end-users to reach refined and transparent disaster-related social media data. At the entry point, data are collected by sending real-time requests through the APIs of platforms such as Twitter, Instagram, YouTube, TikTok, etc. Four distinct models interpret data from social media platforms. These models are tailored to focus on different analysis domains, enabling in-depth examination of the data. At first, a model is used to conduct sentiment analysis to determine the emotional tone of texts, thus assessing the emotional intensity of the content. Then, another model identifies content related to disaster events and determines whether a text mentions a disaster event and, if so, employs a sub-model to ascertain the specific type of disaster, such as earthquake, flood, or fire. Lastly, a model is to discern whether content discusses political, religious, or voluntary themes. By analysing the content of texts, this model identifies prominent themes, thereby determining the primary focus of the content. Afterwards, these models are exposed as APIs, allowing users to interact with them programmatically. Within a .NET project, Hangfire is utilised to send real-time requests, enabling the interpretation of incoming textual data and indexing it into Elasticsearch. MongoDB logs data from the social media APIs in our project because of its flexibility, scalability, performance, querying capabilities, and integration options. Analytical results are



visualised like charts and graphs for straightforward interpretation and could be called an aggregation service by developing dashboard interfaces using the Kibana tool. Kibana's flexibility allows the creation of visually engaging and informative dashboards, enabling users to explore and interact with data seamlessly. To safeguard the privacy of the collected data, measures were taken to anonymise or remove all personally identifiable information (such as names, email addresses, phone numbers, etc.) before utilising it for training purposes. Anonymising the data ensures that individuals cannot be identified from the dataset, thereby protecting their privacy. These practices reflect our commitment to preserving user privacy and conducting the data collection process ethically and legally.

3.13 Ontology-based Knowledge Graph

Ontology is pivotal in representing data with semantics, information and knowledge, standardising terminology and achieving semantic interoperability. It provides an infrastructure to model multi-hazard and multi-hazard-related risks and manage such risks. By providing a structured framework for integration, interdisciplinary collaboration, risk assessment, and decision support, ontology-based knowledge graphs can empower stakeholders at all levels to navigate the complexities of an ever-changing environment in the face of multi-hazard challenges. A conceptual framework was created by structuring the flood ontology into five different facets, defined by considering key aspects: i) event aspect, ii) impact and effect aspect, iii) warning aspect, iv) flood protection aspect, and v) forecast aspect. The expansion of this conceptual framework keeps the flood ontology at its centre, enabling the logical description and data representation of cascading effects of floods, such as landslides, causes of floods, such as earthquakes, and the co-occurrence of other hazards, including wildfires and heatwaves with floods. The generic model of flood represents the flood types, while the event aspect represents spatiotemporal characteristics, and causative factors, providing a comprehensive framework for understanding and managing flood risks. The impact and effect aspect of the flood ontology delves into the consequences of flood events. The warning aspect represents different warning types, capturing historical data and ensuring precise spatiotemporal communication; thus empowering communities to be better prepared for flood events. Flood protection represents the diverse range of structural measures employed to prevent floods. By systematically representing the environmental and geographical factors influencing flood events, the forecast aspect facilitates the development of flood prediction models that adapt to real-time data. Constructing a Virtual Knowledge Graph (VKG) eliminates the requirement for explicit instantiation of concepts for generating triples, unlike traditional knowledge graphs that persistently store triples. The VKG applies a mapping between ontologies and data and produces a unified and virtual representation of the knowledge graph with IRIs of instances formed based on the schema defined in the mapping. By leveraging the unified ontology, the VKG-based federated approach harmonises data from different sources, ensures interoperability, and enables seamless integration using Ontop. Integrating the flood ontology with the VKG has substantial implications for decision support and resilience enhancement. By providing an integrated view of flood data, the VKG equips decision-makers with timely information for risk assessment, emergency response planning, and infrastructure development, ultimately enhancing community resilience.

3.14 Decision Support Tool (DST) for Policy Management

Developing effective policies to tackle natural disasters, including floods, wildfires and earthquakes, is a critical global imperative for policymakers. Despite the wealth of data available to monitor climate change and its impacts on society, the environment, and the economy, policy formulation often relies on intuition rather than evidence-based methodologies, underscoring the need for innovative decision support systems. This endeavour seeks to introduce a decision-support system to guide policymakers through a structured, data-driven decision-making process to address the complexities of managing various natural disasters, including floods, wildfires and earthquakes. It is a multifaceted platform, facilitating policymakers through five phases of the policy cycle: inform, advise, monitor, evaluate, and revise. In each phase, dedicated dashboards furnish policymakers with pertinent information about



environmental, social, and economic conditions, enabling a holistic understanding of the challenges. The decision support system empowers policymakers to make informed decisions at every juncture by harnessing a wealth of data sources and employing advanced analytics. To showcase the efficacy of this approach, the decision support system was applied to assess a comprehensive disaster management policy firstly in the city of Ordu, Turkey, to be followed by the other CSAs of C2IMPRESS: Egaleo, Balearic Islands and Port Averio. Through rigorous analysis, the system unveils the benefits and challenges inherent in the proposed policy, providing invaluable insights for public administrations engaged in disaster risk management endeavours. In essence, this innovative decision support system heralds a paradigm shift in policymaking, ushering in an era of evidence-based governance in disaster risk management. By amalgamating data-driven insights with stakeholder engagement, it offers a robust framework for crafting resilient policies capable of mitigating the impacts of natural disasters on communities, infrastructure, and the environment. Concerning the need for and importance of developing evidence-based and data-validated decision-making processes, the Decision Support Tool (DST) for Policy Management has been designed to support policymakers.

3.15 Multi-Criteria Analysis Tool (MCA)

An interactive multi-criteria analysis tool (MCA) was developed to extract information from the C2IMPRESS microservices representing technical, environmental, social, health and economic criteria and provide a ranking of the schemes and options. It is a decision-making framework that evaluates and prioritises different scenarios or alternatives based on multiple criteria. It is beneficial in complex decision-making processes where various qualitative and quantitative factors must be considered. The MCA tool allows stakeholders to assign weights to different criteria, reflecting their relative importance, and then evaluates the alternatives accordingly. The tool uses techniques like the Analytical Hierarchy Process (AHP) to break down complex decisions into a hierarchy of criteria and sub-criteria, making the decision process more structured and transparent. This tool identifies the decision problem, where the user clearly defines the goals and objectives. This step sets the foundation for determining which aspects of the problem must be addressed. Evaluation criteria are then described, representing the factors relevant to the decision. These criteria must be measurable and aligned with the decision-makers' concerns. The tool allows one to select existing criteria from a database or introduce new ones while allowing for a hierarchical grouping. Scenarios are defined by entering descriptions in the tool and assigning criteria values at the lowest level of the hierarchy. If the values are not readily available, pairwise comparison can be used to estimate them. When criteria have different units of measurement, the MCA tool normalizes the values using methods like linear max, linear min, linear (max-min), and vector normalization to ensure comparability. The importance of each criterion is reflected by assigning weights, which can be done either through direct weighting or pairwise comparison. Afterwards, the tool overlays these weights onto the criteria using a weighted linear method, ranking the scenarios based on the overall scores. The outcomes are presented in tabular and chart formats, displaying scenario-specific scores and criteria weights, providing a clear visual understanding of the analysis. The outcome is ranking scenarios based on their performance against the chosen criteria, allowing users to make more informed decisions. This interactive tool is open access, designed to be user-friendly, and supports collaborative decision-making across various sectors.

3.16 Disaster Monitoring Portal (DMOP)

Through innovative technology and strategic planning, DMOP aims to streamline disaster monitoring, response coordination, and community involvement, ultimately fostering resilience and reducing the impact of disasters on affected populations. It is designed with three primary modules – Weather Monitoring, Disaster Damage & Losses Tracking, and Citizens Forum. The Weather Monitoring module provides real-time monitoring and analysis of meteorological data, including temperature, precipitation, wind patterns, and atmospheric pressure. The weather monitoring module comprises four features: Multi-Hazard Risk Cycles Analysis, Early Warning System, Forecasting and Risk



awareness. For multi-hazard risk cycle analysis, CLIMADA, an open access fully probabilistic risk assessment model, was used to identify periods of heightened risk by analysing historical data of different natural disasters, allowing for proactive mitigation and preparedness measures. The Early Warning System module collects and disseminates warning data, dependent on webhook service to receive data from external providers or sources. For the forecasting module, OpenWeather will be used to fetch data from the OpenWeather server, and this module consists of two main filtering options: filter by disasters and filter by region. Risk Awareness aims to equip individuals and communities with the knowledge and skills to mitigate risks and effectively respond to disasters. This module has two segments: Questions and Answers (Q&A) and First Aid steps, which might be necessary during or after the disaster. The Disaster Damage and Losses Tracking module facilitates systematically recording and assessing disaster-related damages to infrastructure, properties, and livelihoods. For this, GDACS will fetch the list of disasters that are currently happening and have recently happened. The Citizens Forum module serves as a platform for community engagement and collaboration in disaster management. For community engagement in the platform, social media activity is included from different social platforms regarding any disaster happening nearby. A webhook service receives data from external data providers or sources. This tool is also integrated with the C2IMPRESS Central Database, where the integration with the legacy database (Copernicus Climate Change Service (C3S)) is developed. C3S is a legacy database that is vital in providing a wealth of data spanning diverse domains, including climate, atmosphere, land, and others. Furthermore, using machine learning algorithms in virtual disaster analysis extends beyond simple image storage to enhance the platform's capabilities. These algorithms are designed to analyse the content of uploaded images in a sophisticated manner, extracting valuable information about the nature and severity of flood hazards, such as water levels and depth of inundation. It empowers citizens to contribute helpful information and provides decision-makers with actionable insights to improve disaster preparedness, response, and recovery efforts.

3.17 Integrated DPSIR and PESTLE Tool

The DPSIR framework is valuable for assessing disaster risk and resilience by examining five key components: Drivers, Pressures, State, Impact, and Response. This tool allows the user to select a driver and corresponding pressure, assign a weight, and add a PESTLE analysis optionally. The PESTLE analysis section lists Political, Economic, Social, technological, Legal, and Environmental components. In this method, multiple impacts are linked to a single state, multiple states are tied to one pressure, and several pressures stem from one driver. Responses can address drivers, pressures, states, or impacts. In the DPSIR tool, users assign severity scores to each component. To calculate an individual state's Effective Score (ES), the sum of connected impact severity scores is multiplied by the state's user-assigned severity score. Similarly, Pressure and Driver Effective Scores are calculated by summing and multiplying relevant state and pressure scores. The assessment begins by defining the scope and objectives and identifying relevant hazards, vulnerabilities, and risks. The Drivers, which are the underlying causes of disaster risk, can include natural and human-induced hazards, while the Pressures are external factors such as demographic or environmental influences. The State refers to the current physical, social, economic, and ecological conditions that affect disaster risk, while the Impact measures the human, financial, and environmental losses from past disasters. The Response component looks at the actions taken to mitigate risks before, during, and after disasters. Once the assessment is complete, the findings are analysed to identify critical areas of risk and vulnerability, leading to the development of an action plan. This plan aims to reduce disaster risk and enhance resilience through improved early warning systems, strengthened infrastructure, and increased community preparedness.

3.18 The end users, decision support and the integration status of the microservices:

The following table provides a brief summary of the tools, their end users, the decision support they provide and the status of the development.



C2IMPRESS Microservices	Questionnaires	Descriptions
Ontology-Based Knowledge Graph	Target End users	Researchers, academicians, decision makers
	Output for the End users	Interoperable data
	Confidentiality (data, output, and code)	Open source
	Integration to the Platform	Seamless integration
	How the end user will use the tool in the platform	There will be a user interface to use the tool
	Status of the Integration	Ongoing (75% finished)
	Deliverable where the details of the microservices are explained	D4.1
	Main consortium partner	UCAM
MCA	Target End users	Decision makers, researchers, academicians, students
	Output for the End users	Decision support to choose the best alternative or options
	Confidentiality (data, output, and code)	The tool is open access.
	Integration to the Platform	Seamless integration
	How the end user will use the tool in the platform	Through the toll user interface.
	Status of the Integration	Ongoing, (75% done)
	Deliverable where the details of the microservices are explained	D4.2 (section 4)
	Main consortium partner	TVS
SMCDA	Target End users	Decision makers, researchers, academicians, students
	Output for the End users	Risk map with classification

	Confidentiality (data, output, and code)	The tool will be open access
	Integration to the Platform	Seamless integration
	How the end user will use the tool in the platform	Using the C2IMPRESS platform. GUI will be provided.
	Status of the Integration	Ongoing
	Deliverable where the details of the microservices are explained	D3.2
	Main consortium partner	TVS
HorizonActionEye	Target End users	The flexibility of the Microservice has multisectoral targets - ranging from a specialized user base including academic and industry researchers, developers and practitioners, to a broader user base including decision makers, authorities, planners and responders in civil protection, security and emergency forces. Notwithstanding the complexity of the underlying technologies, the end-user outputs will further cater for the interested citizen, namely in raising awareness, better understanding and in providing actionable insights, since ultimately every citizen is a civil protection agent in their own lives and those of their own family and community.
	Output for the End users	Geophysical flows and associated spatiotemporally distributed risks.
	Confidentiality (data, output, and code)	The overall Microservice is Sensitive (SEN).
	Integration to the Platform	The tool will be hyperlinked to the platform, so that it can be seamlessly accessible from there whilst retaining its functional autonomy.
	How the end user will use the tool in the platform	The user can consult, explore and retrieve/download the results produced in the platform, along with the associated analytics.
	Status of the Integration	The main tool development is ongoing.
	Deliverable where the details of the microservices are explained	D4.1

	Main consortium partner	MET
Big Data-Powered Decision Support Tool	Target End users	The primary end users are policymakers and public administrators responsible for disaster risk management, such as government agencies at local, regional, and national levels. These users are involved in managing risks from natural disasters.
	Output for the End users	• A structured, workflow-driven decision-support system that assists end users through all phases of the policy cycle (inform, advise, monitor, evaluate, revise). • Dashboards that provide real-time, social media data-driven insights into disaster events. • Advanced analyses that help evaluate and follow different policy options based on disaster risks, offering clear comparisons for decision-making. • Practical case study outputs, such as from the city of Ordu, Turkey, that showcase real-world application of the tool.
	Confidentiality (data, output, and code)	Data collected and used within the system will be handled according to data privacy regulations, ensuring sensitive information is protected. The outputs generated by the tool, such as policy recommendations and evaluations, will be made accessible only to authorized users, while if needed, public-facing reports will contain anonymized or non-sensitive data. The underlying processes and system code are proprietary, but can be shared or accessed by authorized stakeholders within the framework of the project.
	Integration to the Platform	The BPM platform has the capacity to be seamlessly integrated into the C2IMPRESS platform, allowing users to see the results of the workflows of DST tool (T4.5) within the T4.7 platform. Through embedded components, such as widgets and a dashboard interface, users will be able to display policy-making workflows and view results without leaving the C2IMPRESS platform. The integration will enable users to see the policy decision-making processes' results ensuring a smooth, cohesive experience for decision-makers.
	How the end user will use the tool in the platform	End users will see the specific and public results of the DST tool through an interface that provides access to the DST platform's dashboards that is integrated to the T4.7 platform. The system enables collaboration and feedback within the policy formulation and disaster management processes.
	Status of the Integration	The main tool development is complete (t4.5). The details on the platform is on progress.
	Deliverable where the details	D4.1

	of the microservices are explained	
	Main consortium partner	SAMPAS
DPSIR and PESTLE	Target End users	The authorities who are interested in causal chain analysis and willing to obtain mitigation measures as response, policy makers, academicians.
	Output for the End users	Causal chain of a system and response.
	Confidentiality (data, output, and code)	The toll is open access.
	Integration to the Platform	Seamless integration
	How the end user will use the tool in the platform	Using the C2IMPRESS platform.
	Status of the Integration	Done.
	Deliverable where the details of the microservices are explained	D1.1 and D4.2
	Main consortium partner	TVS
HIDRALERTA	Target End users	Port authorities, Municipality officials, Civil protection officials
	Output for the End users	3-day wave forecasting in a range of cascading spatial domains (from the ocean basin scale to the harbor basin scale), 3-day port structures overtopping forecasting, 3-day ship's UKC forecasting and 3-day ship's mooring forces forecasting. Maps, alert bulletins, alert tables, alert emails and parameters graphs.
	Confidentiality (data, output, and code)	Data and code is confidential. Outputs are open
	Integration to the Platform	Output Image maps (png), alert bulletins (pdf) and parameters (csv) will be provided to embed the tool in the platform.
	How the end user will use the tool in the platform	The user can view the generated results and can perform some graphical analysis, using the numeric output data
	Status of the Integration	yet to start; the main tool development is ongoing
	Deliverable where the details	D4.1

	of the microservices are explained	
	Main consortium partner	LNEC
Fluvial Flooding and Landslide in the Ordu, Turkey	Target End users	Authorities (Ordu Metropolitan Municipality)
	Output for the End users	Forecasting for flood risk in a day and warning for authorities for deciding measurements
	Confidentiality (data, output, and code)	Open access to authorities.
	Integration to the Platform	Not decided yet. Need to be discussed.
	How the end user will use the tool in the platform	the user can perform analysis, upload data, and get back analysis results
	Status of the Integration	Yet to start
	Deliverable where the details of the microservices are explained	D1.2, D4.1
	Main consortium partner	ORDU
Flood Hazards in the Mallorca, Balearic Island	Target End users	Citizens, academics, authorities, decision-makers, and policymakers
	Output for the End users	Observed data in real time (water level (5 min), water temperature (5 min), air temperature (10 min), relative humidity (10 min), soil temperature (30 min), soil humidity (10 min) and accumulated precipitation (10 min) and simulated data from hydrological modelling (3 warning levels (yellow, orange and red) for bridges and fords. Also, simulated data from hydraulic modelling as water velocity and water stage raster's and flooding areas.
	Confidentiality (data, output, and code)	Open access
	Integration to the Platform	The forecasting tool is developed and will be integrated with the Platform (T4.7) via I-frame.
	How the end user will use the tool in the platform	The user can only view the generated results; the user can perform analysis, upload data, and get back analysis results, etc.
	Status of the Integration	The main tool development is ongoing. As soon as it is finished the I-frame will be provided to TVS for integration

	Deliverable where the details of the microservices are explained	D1.3
	Main consortium partner	UIB and GOIB
Wildfire Hazards in the Balearic Island	Target End users	Citizens, academics, authorities, decision-makers, and policymakers
	Output for the End users	Daily mapping of weather indicators (i.e., 30/30/30, Fire Weather Index); Weekly cartography updated at municipal scale and by forest masses on the current state of the vegetation; Daily bulletins predicting the probability of occurrence of IF by forest stands and municipalities; Vulnerability study of the risk of forest fires in the Balearic Islands.
	Confidentiality (data, output, and code)	Github, Open access
	Integration to the Platform	Via I-frame.
	How the end user will use the tool in the platform	The user can only view the generated results
	Status of the Integration	Ongoing/Complete/yet to start; the main tool development is ongoing. The I-frame will be provided later.
	Deliverable where the details of the microservices are explained	D1.2
	Main consortium partner	UIB and GOIB
	Target End users	The generality and flexibility of the framework has multisectoral targets - ranging from a specialized user base including academic and industry researchers, developers and practitioners, to a broader user base including decision makers, authorities, planners and responders in civil protection, security and emergency forces. Notwithstanding the complexity of the underlying technologies, the end-user outputs will further cater for the interested citizen, namely in raising awareness, better understanding and in providing actionable insights, since ultimately every citizen is a civil protection agent in their own lives and those of their own family and community.
SoS4MHRIN		

	Output for the End users	The output catalogue is still being expanded. The current list is thus not final. Outputs include ESDI metocean forecasts and analytics, covering Europe at different available spatiotemporal scale, resolution and lead settings, along with hazard and multi-hazard metrics in space and time, including pertaining the interplay among e.g. pyro, weather, hydrologic and other mechanisms shaping a multivariate spatiotemporal multi-hazard nexus and associated mechanisms. Final product offerings are subject to change as the tool evolves with new functionalities being incorporated and as further ESDI models mature and become validated for operational use.
	Confidentiality (data, output, and code)	SoS4MHRIN is sensitive (SEN) and shall be available through licensing.
	Integration to the Platform	The tool will be hyperlinked to the platform, so that it can be seamlessly accessible from there whilst retaining its functional autonomy.
	How the end user will use the tool in the platform	The user can consult, explore and retrieve/download the results produced in the platform, along with the associated analytics.
	Status of the Integration	The main tool development is ongoing, along with the production and validation of scientific and technical content to be provided in the final platform. The initial planning on the integration has been done.
	Deliverable where the details of the microservices are explained	D2.3
	Main consortium partner	MET
ABM & HBM	Target End users	Emergency planners, local authorities, decision-makers, and policymakers
	Output for the End users	Egaleo (earthquakes): output of 3 scenarios (1. baseline that covers current evacuation policy; 2. Policy change so that evacuation centres are pre-assigned to individuals; 3. Scenario 2 plus human behaviour model that captures real-world behaviours of individuals. Mallorca (Floods) output of 4 scenarios (1. baseline that covers current evacuation policy; 2. Baseline plus integration of an early-warning system (EWS); 3. Human behaviour without EWS; 4. Human behaviour with EWS). Mallorca (Forest fires) output of 3 scenarios.

	Confidentiality (data, output, and code)	Sensitive
	Integration to the Platform	Selected output will be displayed in the platform. The ABM and HBM model is specialised model and thus one should request to the partner for further application.
	How the end user will use the tool in the platform	The user can view the generated results of all of the scenarios plus their analyses (e.g. number of people in each evacuation centre for each scenario, congestion of roads for each scenario, status of people during flood scenarios (e.g. at home, seeking shelter, stuck, moving to higher ground or to an evacuation area)).
	Status of the Integration	The main tool development is ongoing. Initial planning for integration has been done.
	Deliverable where the details of the microservices are explained	D4.2
	Main consortium partner	RMIT-EU, RMIT and UGA
DMOP	Target End users	For all types of users (citizens, academicians, managers etc)
	Output for the End users	Disaster information all over the world and certain decision support.
	Confidentiality (data, output, and code)	Open access
	Integration to the Platform	Seamless integration
	How the end user will use the tool in the platform	Through the C2IMPRESS platform (T4.7)
	Status of the Integration	Done
	Deliverable where the details of the microservices are explained	D4.1
	Main consortium partner	TVS
Risk & Resilience Framework	Target End users	Local and regional administrations, civic society / NGO, civil protection organizations, and policy-makers

	Output for the End users	• Hazard levels / likelihood for specific hazard levels • Exposure maps to specific hazard levels • Risk levels for specific multi-hazard scenarios (map scale) • Resilience levels (spider plot)
	Confidentiality (data, output, and code)	Most appropriate if those who enter data select confidentiality level. (e.g. open, access for selected groups, only for data providers)
	Integration to the Platform	Via a web app where users can select data
	How the end user will use the tool in the platform	Perform analysis on the tool
	Status of the Integration	The main tool development on-going. The planning will be done soon.
	Deliverable where the details of the microservices are explained	D3.4
	Main consortium partner	NCSR
PPCP	Target End users	Academics, authorities, decision-makers, policymakers, and civil society.
	Output for the End users	Polycentric Governance Framework adapted to each CSA (thanks to specific PPCP tools as questionnaire, stakeholders mapping, SWOT analysis).
	Confidentiality (data, output, and code)	not relevant.
	Integration to the Platform	The PPCP tools and the webinar will be available for the users. The methodological guideline will be included in the platform.
	How the end user will use the tool in the platform	The user can watch the webinar and can experiment the PPCP tools. Also the user can download the general method.
	Status of the Integration	Ongoing (elements have to be submitted to the platform)
	Deliverable where the details of the microservices are explained	D7.1
	Main consortium partner	ARTELIA
social media analytics	Target End users	Decision-makers, policymakers, authorities, researchers, and emergency management professionals who need real-time data insights on disaster-related events from social media platforms.

	Output for the End users	• Real-time social media sentiment analysis related to disasters. • Identification of disaster-related content (e.g., earthquake, flood, wildfire). • Thematic analysis of posts (e.g., political, religious, or voluntary aspects). • Visualized analytical results in the form of charts, graphs, and dashboards to track and interpret disaster trends.
	Confidentiality (data, output, and code)	Data is anonymized to protect user privacy, with all personally identifiable information (such as names, emails, phone numbers) removed before analysis. The output, data, and code are handled confidentially, adhering to ethical and legal guidelines to ensure privacy protection.
	Integration to the Platform	Widgets and interactive dashboards will be provided to embed the tool within the C2IMPRESS platform (T4.7). Users can view and interact with the visualized analytical results directly on the platform, which seamlessly integrates social media data analysis with disaster management workflows.
	How the end user will use the tool in the platform	The end user can perform analysis by filtering specific criteria or events, or filter in different time interval and receive analysis results in real-time. They can also view and interact with the pre-generated results through visual dashboards and charts.
	Status of the Integration	The tool development is complete. The widgets will be provided for integration in T4.7. Ongoing.
	Deliverable where the details of the microservices are explained	D4.1
	Main consortium partner	SAMPAS
FlowBasin	Target End users	Any person interested in delineate and characterize watersheds and have a first evaluation of the discharge under certain conditions. This may include water management decision-makers, academics, engineers or citizens in general.
	Output for the End users	The output will be the watershed shape (which can be downloaded to be used in GIS for example), a report about the watershed characteristics (area, perimeter, hypsometric curve, etc) and the estimation of the flow discharge for several return periods. Inundation areas given by the Height Above Nearest Drainage (HAND) methodology will also be obtained.

	Confidentiality (data, output, and code)	Code will be confidential. The tool will be open access.
	Integration to the Platform	An I-frame will be provided to embed the tool in the platform.
	How the end user will use the tool in the platform	The user can perform analysis, upload data, and get back analysis results.
	Status of the Integration	Yet to start; the main tool development is ongoing.
	Deliverable where the details of the microservices are explained	D6.3
	Main consortium partner	LNEC

4 Development of Decision Support Tools

As mentioned in the proposal, TVS has developed a multicriteria decision analysis (MCDA) tool and another decision support tool – Integrated DPSIR and PESTLE analysis which has been formulated in WP1. This section provides an overview of these tools and some features with screenshots. Both tools are fully developed.

4.1 Multi-Criteria Analysis Tool (MCA)

4.1.1 Introduction

The Multi-Criteria Analysis (MCA) tool is a decision-making framework used to evaluate and prioritize different scenarios or alternatives based on multiple criteria [10]. It is beneficial in complex decision-making processes where various qualitative and quantitative factors must be considered. The MCA tool allows stakeholders to assign weights to different criteria, reflecting their relative importance, and then evaluates the alternatives accordingly [2]. In the context of the tool being developed, the MCA method helps rank or prioritize scenarios by integrating inputs from experts and citizens. The tool uses techniques like the Analytical Hierarchy Process (AHP) to break down complex decisions into a hierarchy of criteria and sub-criteria, making the decision process more structured and transparent. The outcome is ranking scenarios based on their performance against the chosen criteria, allowing users to make more informed decisions [9]. This interactive tool is open access, designed to be user-friendly, and supports collaborative decision-making across various sectors.

4.1.2 Concept

The Multi-Criteria Analysis (MCA) tool helps prioritize scenarios by evaluating them across various criteria. The process starts by identifying the decision problem, where the user clearly defines the goals and objectives [1]. This step sets the foundation for determining which aspects of the problem must be addressed. Evaluation criteria are then defined, representing the factors relevant to the decision. These criteria must be measurable and aligned with the decision-makers' concerns. The tool allows one to select existing criteria from a database or introduce new ones while allowing for a hierarchical grouping.

Scenarios are defined by entering descriptions in the tool and assigning criteria values at the lowest level of the hierarchy. If the values are not readily available, pairwise comparison, currently under



development, can be used to estimate them. When criteria have different units of measurement, the MCA tool normalizes the values using methods like linear max, linear min, linear (max-min), and vector normalization to ensure comparability [3].

The importance of each criterion is reflected by assigning weights, which can be done either through direct weighting or pairwise comparison. Afterward, the tool overlays these weights onto the criteria using a weighted linear method, ranking the scenarios based on the overall scores [4]. The outcomes are presented in tabular and chart formats, displaying scenario-specific scores and criteria weights, providing a clear visual understanding of the analysis.

4.1.3 Interfaces & Examples

Development of Common Interfaces

This cover page highlights the tool's ability to evaluate multiple alternatives or scenarios against predetermined criteria, helping decision-makers assess various options comprehensively. Key steps in conducting MCA, such as identifying the decision problem and defining evaluation criteria, are outlined.



This is a **Multi-Criteria Analysis Tool** which is an effective tool for making decisions because it analyzes and assesses a number of possibilities or alternatives systematically against a set of predetermined criteria.

Multi-Criteria Analysis is a decision-making technique that takes into account multiple criteria or factors when evaluating alternatives or options/scenarios. It helps decision-makers consider various dimensions or perspectives simultaneously, enabling a more comprehensive assessment of the available choices.

Key steps involving in conducting a Multi-Criteria Analysis are as following -

1. **Identify the Decision Problem:** Clearly define the decision problem and the objectives or goals you want to achieve. This step involves understanding the decision context, the stakeholders involved, and the available alternatives.

Figure 4.1: User interface of Multicriteria Analysis Tool

To perform the prioritization, a user will execute the following steps:

Identify the Decision Problem: Clearly define the problem and the objectives or goals you want to achieve.

Define Evaluation Criteria and hierarchy definition: Identify the criteria or factors relevant to the decision problem. These criteria should be measurable and specific and represent the objectives or concerns of the decision-makers. The tool allows the user to select criteria saved in the database. New criteria can be used. The tool also allows us to develop the hierarchy by grouping the criteria and displaying the final hierarchy.

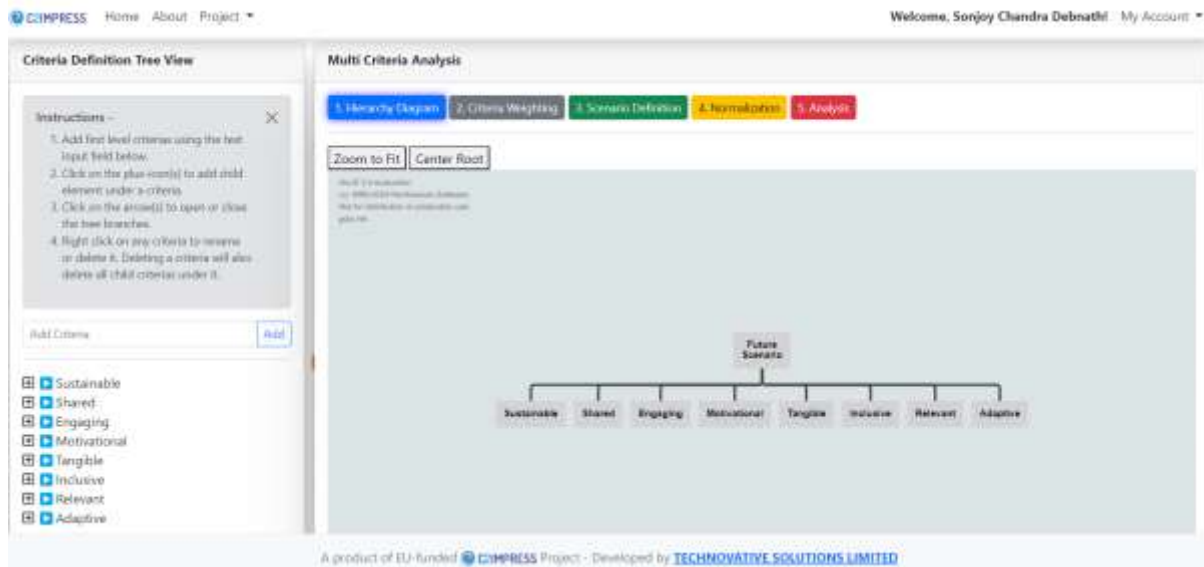


Figure 4.2: An example of an MCA project

This figure is an example of an existing MCA project in our platform. It shows a criteria definition tree view on the left and a hierarchy diagram on the right. The tree view includes criteria such as Sustainable, Shared, Adaptive, etc. The hierarchy diagram visually represents the structure and relationships between the criteria and sub-criteria.

Weight the Criteria: Assign weights or importance values to each criterion to reflect their relative significance in decision-making. We have developed two weighting methods: direct weighting and pairwise comparison. Direct weighting allows the user to provide the weights directly; pairwise comparison allows a method to estimate the weights.

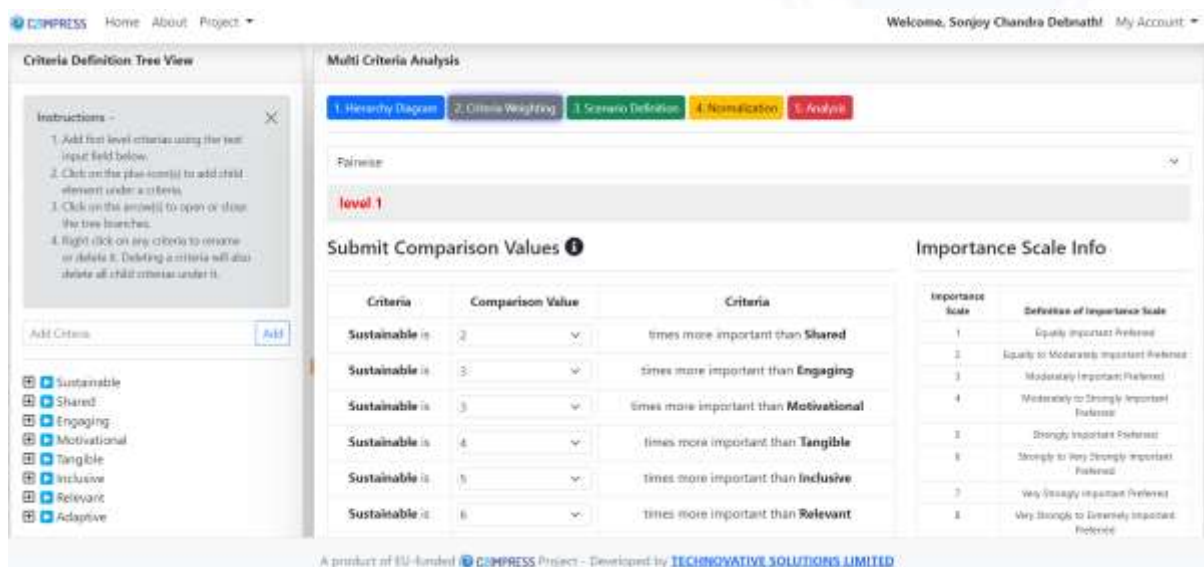


Figure 4.3: An example of using a multi-criteria analysis tool for displaying pair-wise comparisons on the right

Scenario definition: At this stage, the tool allows the definition of the scenarios and the provision of a description in a text box.

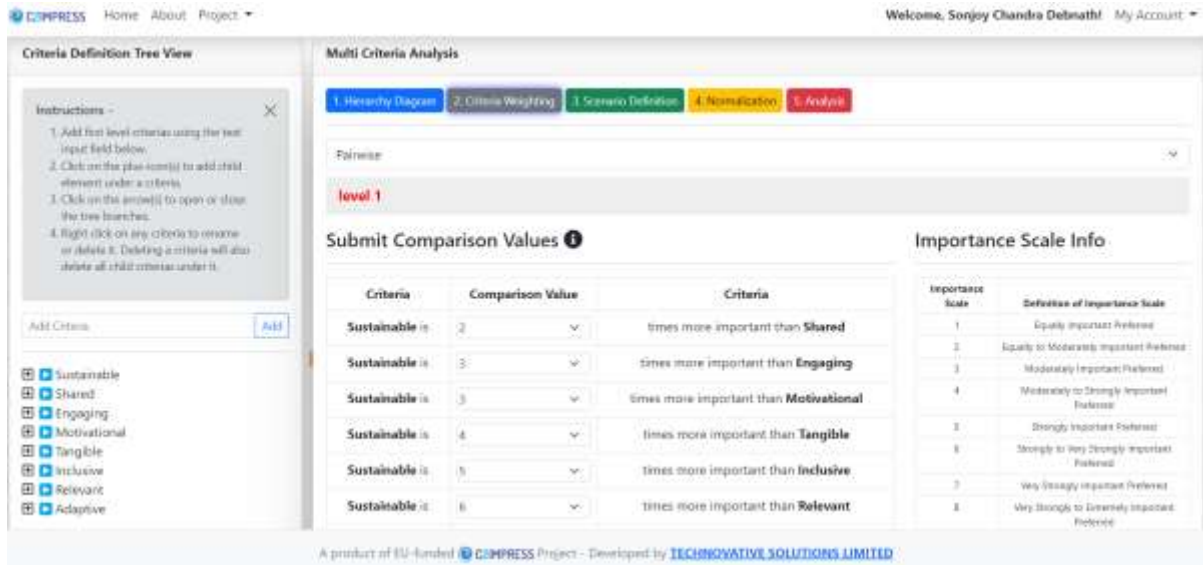


Figure 4.4: A screenshot of an exemplary MCA project

Normalisation of criteria values: The tool normalises the values to make all the values dimensionless. If a pairwise comparison is made, normalisation is not required. If different criteria are measured, normalisation needs to be done. At this stage of development, this tool provides four normalisation methods: linear max, linear min, linear (max-min), and Vector normalisation. In the future, we will include more methods if required.

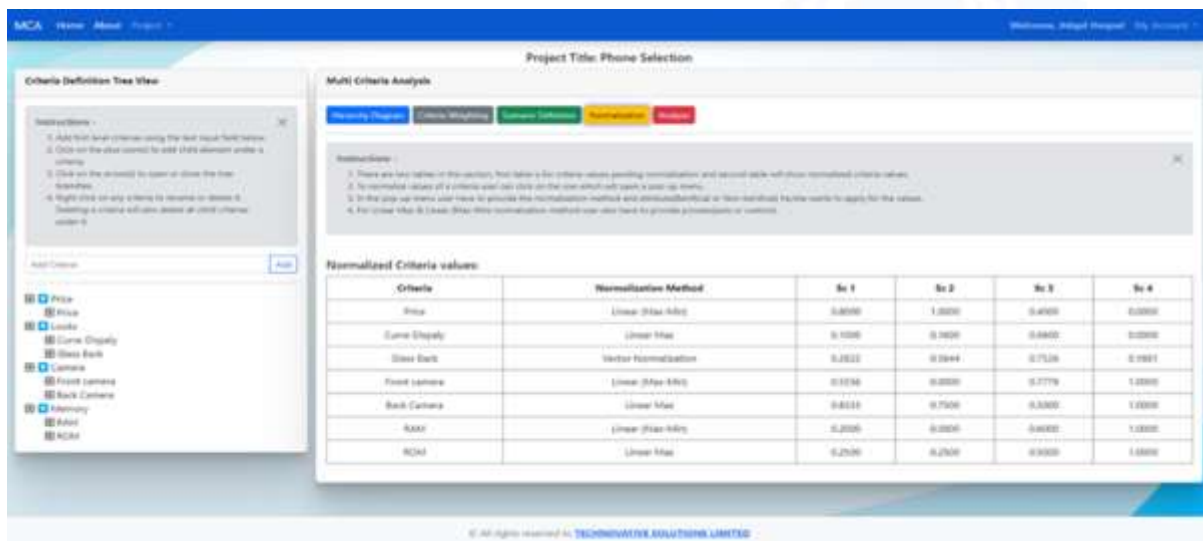


Figure 4.5: An example of normalisation functions

The normalized values table includes criteria across four scenarios, using different normalization methods.

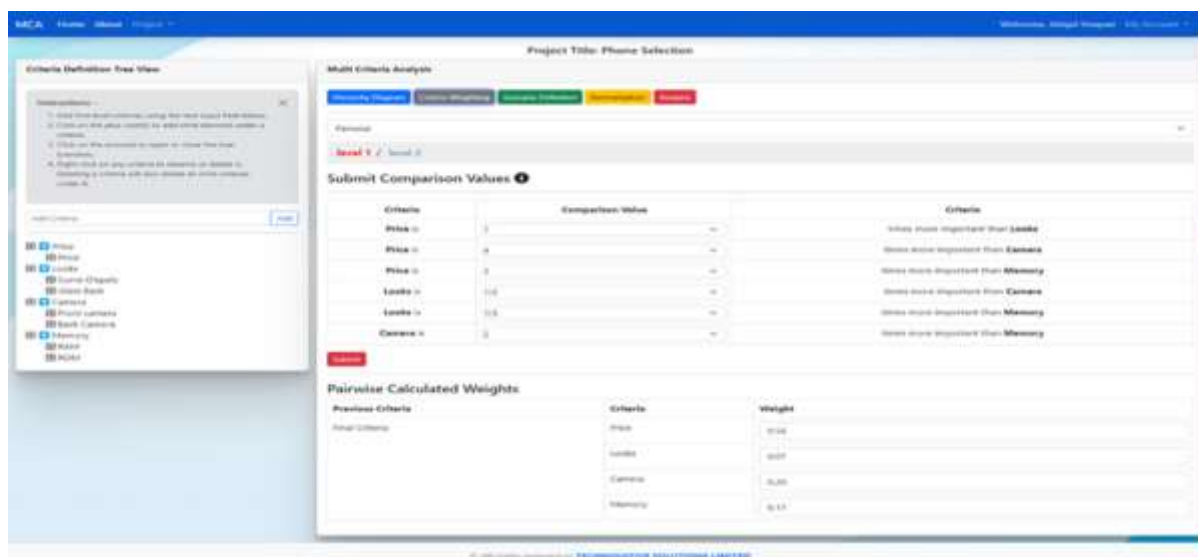


Figure 4.6: A comparison of values submitted for normalisation functions

Analysis section: This section includes values of level 1 criteria with weights and scenario-specific scores for each criterion. The calculated values are displayed for the scenarios. This screenshot displays the final level criteria values after applying weights, the combined values for each scenario, and a bar chart comparing the final ranking of scenarios. The chart shows the percentage scores for scenarios Sc 1, Sc 2, Sc 3, and Sc 4.

In the next version of the MCA, we will include more overlay methods and sensitivity analysis.

After Combination of Final Level Criteria:

Criteria	Sc 1	Sc 1 (%)	Sc 2	Sc 2 (%)	Sc 3	Sc 3 (%)	Sc 4	Sc 4 (%)
Final Ranking	0.2811	28.11	0.3253	32.53	0.1408	14.08	0.2325	23.25

Final Level Criteria vs Scenario After-Combination values in (%)



Figure 4.7: The screenshot displays the final level criteria values

4.2 DPSIR and PESTLE

4.2.1 Introduction

The DPSIR framework is valuable for assessing disaster risk and resilience by examining five key components: Drivers, Pressures, State, Impact, and Response. The assessment begins by defining the



scope and objectives and identifying relevant hazards, vulnerabilities, and risks. The Drivers, which are the underlying causes of disaster risk, can include natural and human-induced hazards, while the Pressures are external factors such as demographic or environmental influences. The State refers to the current physical, social, economic, and environmental conditions that affect disaster risk, while the Impact measures the human, economic, and environmental losses from past disasters [8]. The Response component looks at the actions taken to mitigate risks before, during, and after disasters. Once the assessment is complete, the findings are analyzed to identify key areas of risk and vulnerability, leading to the development of an action plan. This plan aims to reduce disaster risk and enhance resilience through improved early warning systems, strengthened infrastructure, and increased community preparedness.

4.2.2 Concept

The DPSIR framework connects Drivers, Pressures, States, Impacts, and Responses to assess disaster risks. In this method, multiple impacts are linked to a single state, multiple states are tied to one pressure, and several pressures stem from one driver. Responses can address drivers, pressures, states, or impacts. In the DPSIR tool, users assign severity scores to each component [11]. To calculate an individual state's Effective Score (ES), the sum of connected impact severity scores is multiplied by the state's user-assigned severity score. Similarly, Pressure and Driver Effective Scores are calculated by summing and multiplying relevant state and pressure scores [5].

4.2.3 Interfaces & Examples

Development of common interfaces

The tool provides an interactive user interface to perform the activities. It always provides result analysis graphics for better visualisation. Some of the screenshots are given in the annexe. The tool offers a knowledge base that contains different examples using the DPSIR and PESTLE approaches. One can export outputs in different formats. Some features of the tool are given below:

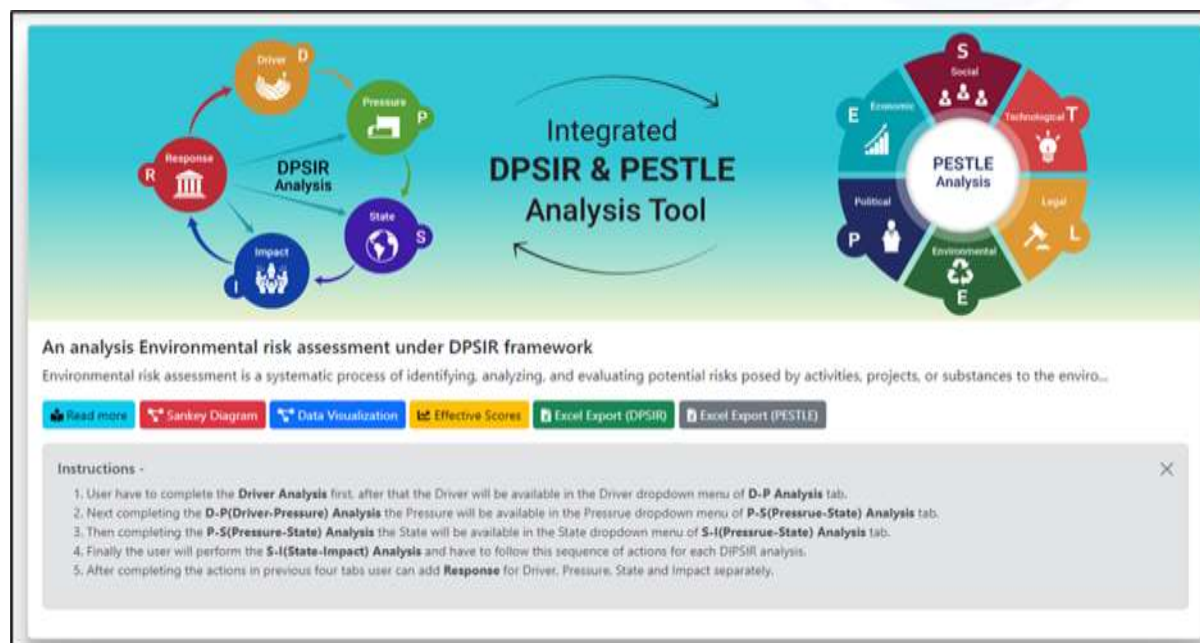
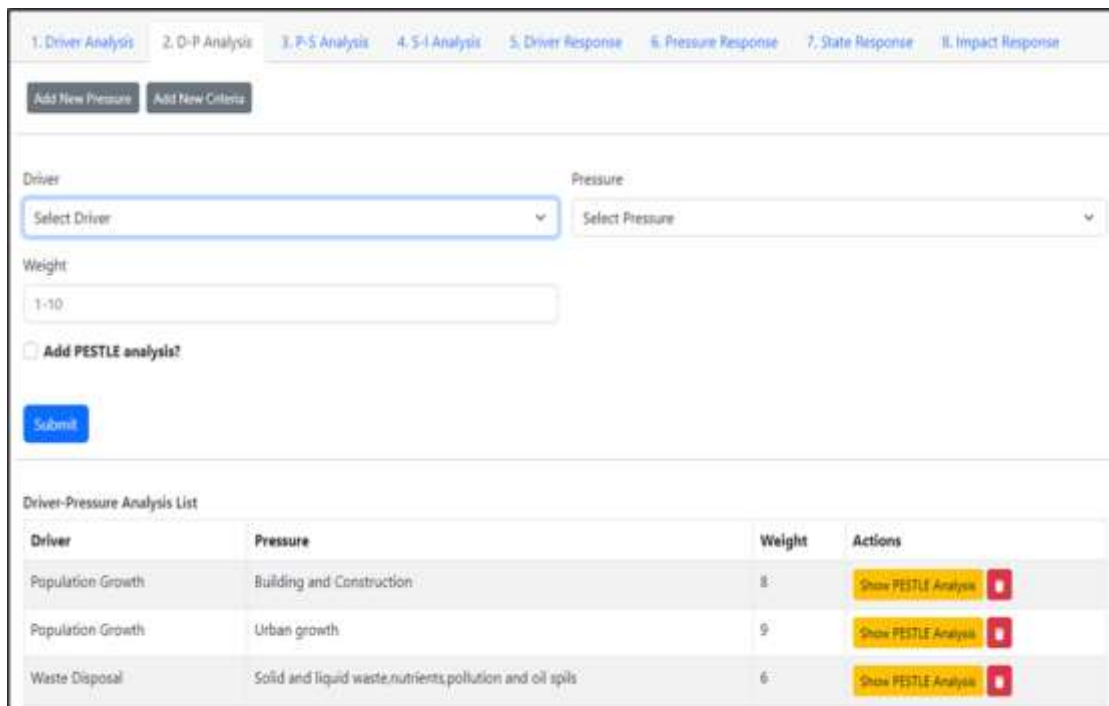


Figure 4.8: Features after login to the project

After logging in to the project, the tool shows relevant analysis features. The features are: Read more, Sankey Diagram, Data Visualisation, Effective score, and Excel Export for DPSIR and DPSIR.

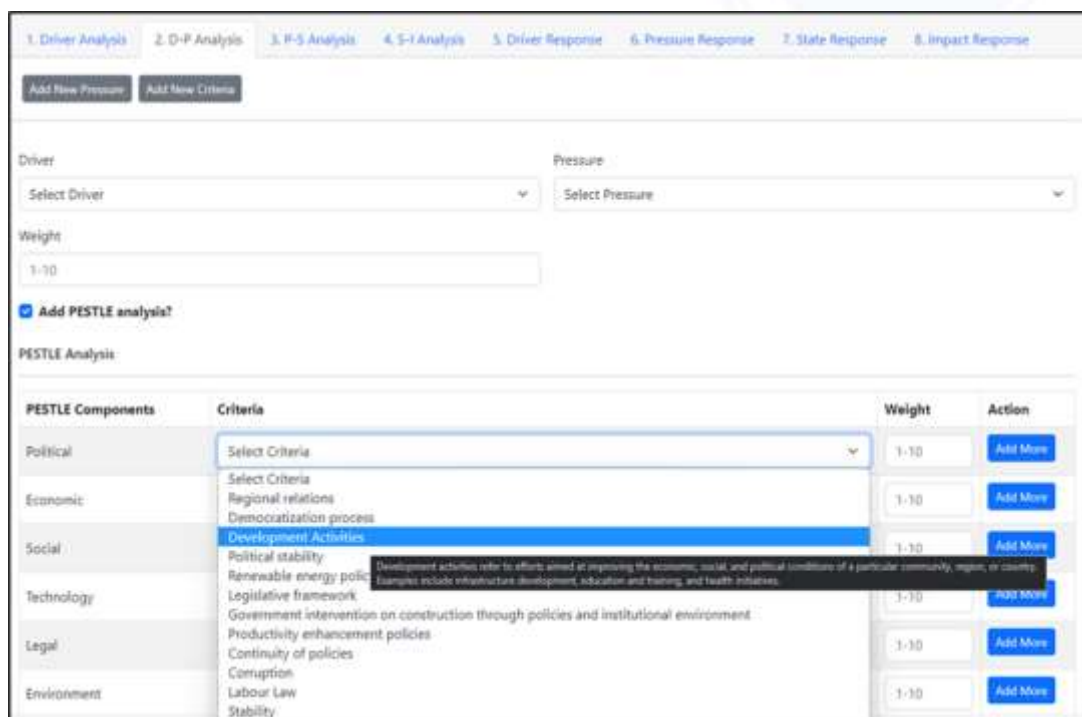


The interface shows a navigation bar with steps 1 to 8. Below it are buttons for 'Add New Pressure' and 'Add New Criteria'. The main form has two dropdown menus for 'Driver' and 'Pressure', a 'Weight' input field (1-10), and a checkbox for 'Add PESTLE analysis?'. A 'Submit' button is at the bottom. Below the form is a 'Driver-Pressure Analysis List' table.

Driver	Pressure	Weight	Actions
Population Growth	Building and Construction	8	Show PESTLE Analysis
Population Growth	Urban growth	9	Show PESTLE Analysis
Waste Disposal	Solid and liquid waste, nutrients, pollution and oil spills	6	Show PESTLE Analysis

Figure 4.9: The interface allows the user to select a driver and corresponding pressure

This interface allows the user to select a driver and corresponding pressure, assign a weight, and add a PESTLE analysis optionally. Below the input fields is a 'Driver-Pressure Analysis List' displaying drivers such as 'Population Growth' and 'Waste Disposal' with their associated pressures, weights, and options to show PESTLE analysis or delete entries.



The interface shows the same navigation bar and form as Figure 4.9, but with the 'Add PESTLE analysis?' checkbox checked. Below the form is a 'PESTLE Analysis' table.

PESTLE Components	Criteria	Weight	Action
Political	Select Criteria	1-10	Add More
Economic	Select Criteria	1-10	Add More
Social	Regional relations	1-10	Add More
Technology	Democratization process	1-10	Add More
Legal	Development Activities	1-10	Add More
Environment	Political stability	1-10	Add More
	Renewable energy policy	1-10	Add More
	Legislative framework	1-10	Add More
	Government intervention on construction through policies and institutional environment	1-10	Add More
	Productivity enhancement policies	1-10	Add More
	Continuity of policies	1-10	Add More
	Corruption	1-10	Add More
	Labour Law	1-10	Add More
	Stability	1-10	Add More

Figure 4.10: The PESTLE analysis section

The PESTLE analysis section lists Political, Economic, Social, technological, Legal, and Environmental components. A dropdown menu allows users to select specific criteria for each component. Users can assign weights and add more criteria for each PESTLE component.

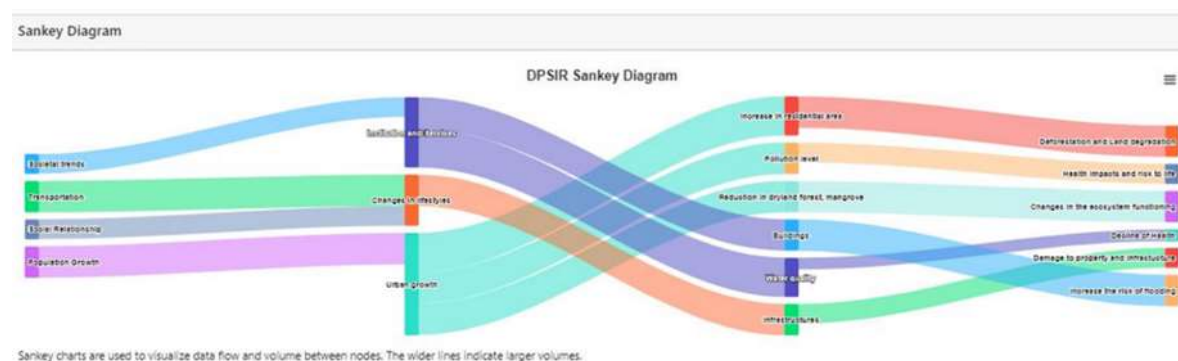


Figure 4.11: Sankey Diagram

Sankey Diagram visualizes data flow and volume between nodes. The diagram shows connections from various drivers, such as 'Population Growth', 'Transportation', and 'Urban growth', to pressures, states, impacts, and responses. More comprehensive lines indicate larger volumes of data flow, illustrating relationships and the magnitude of each connection.

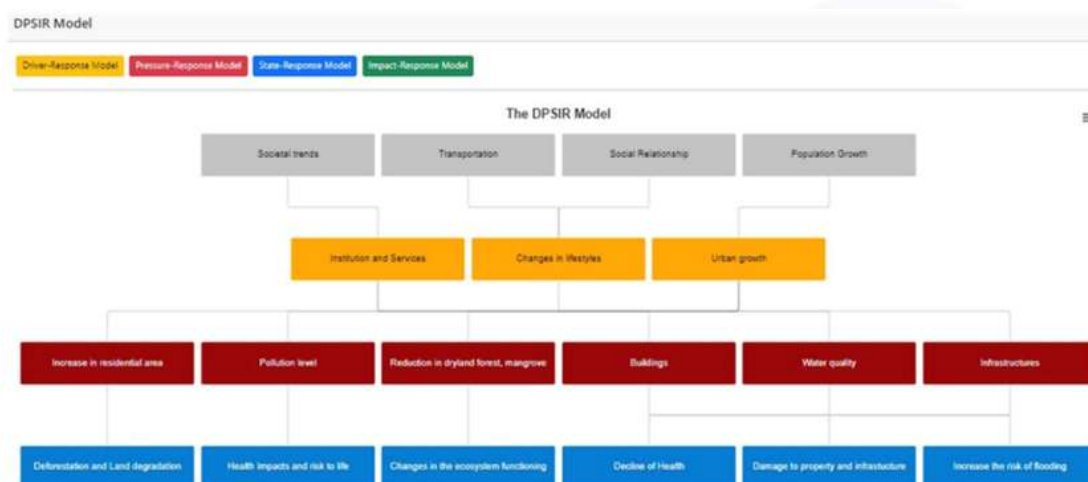


Figure 4.12: Model showing how each element connects within the DPSIR framework

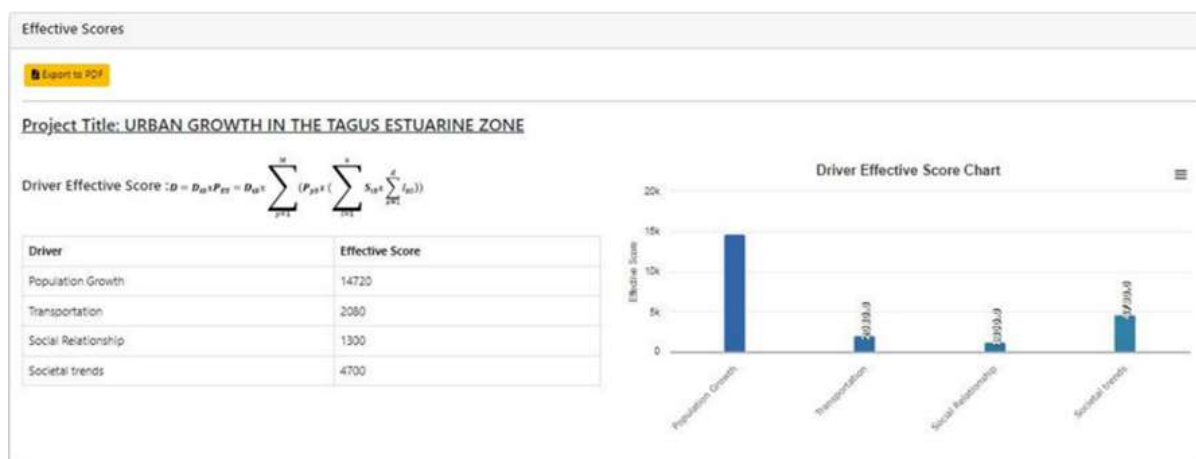


Figure 4.13: Driver effective scores with chart and equation

It Displays a formula for calculating the Driver Effective Score and a table listing drivers and their effective scores. A bar chart visualizes these scores for comparison, and there is an option to export the data to PDF.

5 Next Steps

The integration of the different tools is in progress. As most of the tools in different WPs are underdeveloped, the consortium partners have not provided the tools. As long as they are finished, it will be connected to the platform.

Detail of other tools and their demonstration will be provided in D4.2 v2.

6 References

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